

Capstone 2 - Final Project Report

Florida Polytechnic University

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Project Title: Lakeland Electric Load Modeling

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Course: IDC 4943

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1. Introduction

Lakeland Electric faces challenges forecasting electricity consumption due to constraints of company models that sufficiently model electricity demand and usage. Industry models rarely model local populations' individual, unique, short-term behaviors. To solve this design problem and forecast Lakeland Electric's residential customers' use, our team constructed a unique model for Lakeland Electric residential customers. Our motivation to complete this project came from the desire to generate forecasts that were more accurate, responsive, and transparent than the more exacting industry tools available to Lakeland Electric. For example, Itron uses general models to project electricity usage. Lakeland Electric, the client for this project, is a municipal utility provider who wanted to improve its operational planning and rate setting. The utility wanted to find a forecasting tool that built predictive forecasting tools that tracked residential consumption patterns in 'real time' by using local variables and high-frequency data. Our custom model was constructed to use local inputs such as weather patterns, pricing models, and customer-specific loading behavior to move beyond traditional forecasting solutions to customer operational forecasting. Significant milestones in the design and prototyping included a forecasting equation based on variables such as price, GDP, population, wind speed, heating degree days (HDD), and cooling degree days (CDD). These variables were weighted with the elasticity ranges that emerged from the thorough literature review. The final equation served as the basis for subsequent sensitivity analyses and model assessment. Using multiple rolling window analyses across varying periods (1-month, 1-year, and 10-year windows) allowed us to assess the performance of the model over different forecasting horizons. Bias corrections were also applied chronologically by hour to correct observed load versus predicted load differences. During the course of the project, five different forecasting models were assessed: Research-Based OLS, Lakeland Fit OLS, SARIMAX, Holt-Winters, and Random Forest. This report covers the development, assessment, and finalization of a bespoke electricity load forecasting model for the residential sector of Lakeland Electric. It includes showing relevant research findings, developing a forecasting equation with price, GDP, population, wind speed, heating and cooling degree days, and elasticity ranges from literature; implementing and comparing five forecasting models (research-based OLS, Lakeland Fit OLS, SARIMAX, Holt-Winters, and Random Forest); suggesting sensitivity analysis; and rolling window/hours bias correction. This document only focuses on residential data and short-term forecasting, not commercial or industrial usage or long-term policy planning. This document seeks to illustrate how localized forecasting models can create better operational insights and decision support for a municipal utility like Lakeland Electric.

2. Customer Needs, Objectives, and Constraints

This project will primarily service Lakeland Electric, a municipal utility that provides electricity to the residents in the Lakeland, Florida area. The staff at Lakeland Electric (especially those involved in operations, planning, and rate management) will be the end-

users of the forecasting model. During early project discussions and exploratory research, Lakeland Electric outlined several important needs they have and challenges they face. One of the key pain points they face is that their current forecasting tool, Itron, provides inaccurate and rigid forecasts. While historically sufficient for system-wide forecasting, Itron does not provide granular hourly-level forecasts, nor does it provide duration forecasts for high/low price situations or incorporate short-term demand. Originally, the project identified the need for a model to support real-time revenue and load forecasting for the selection of operational action. However, as the team got into the project in detail and with consideration of the data and technical limitations encountered, the team did not feel that we could really do real-time forecasting due to the granularity of the data (as mentioned), extensions of shaping demand and capacity, and capacity of the utility's current systems and infrastructure. Therefore, the project design shifts the requirements away from real-time forecasting and focuses instead on an hourly or daily load forecasting model (usable aggregate periods) that would still retain all important usage patterning while remaining realistic and executable operationally in consideration of Lakeland Electric's existing operational and infrastructure capabilities. The project scope offered Lakeland Electric, as the utility developed the proposals and design decisions, the scenario to meet their ultimate goals of improving forecast accuracy, informing planning, and preparing for the expense of changing demand. Since I was working to both the organizations' goals when considering inputs from Lakeland Electric and with this relationship, along with the design objectives drawn from a comprehensive understanding of utilities objectives and challenges with forecasting (for objectives in this report, see list of design objectives), the fundamental objective (component of design objectives) was to create a forecasting model that would predict residential electricity load and revenue forecasting using Lakeland electricity data with hourly and daily resolution. The project also needs to design the model to match real customer usage patterns and local environmental conditions, including temperature, wind speed, heating and cooling degree days (HDD/CDD), electricity prices, population estimates, and GDP. The model was designed to be used with three forecast intervals: 1 month, 1 year, and 10 years, to help model both short- and long-term planning periods. Another main purpose of the model was to facilitate scenario analysis of elasticity-based modeling to allow Lakeland Electric to examine the effects of changes in rate or demand increase in load behavior. The results of the model were also expected to be interpretable and allow for the demonstration of patterns of usage and peak demand periods according to several considered conditions. It was important that the system could be used interchangeably, so it was designed with the thought that it would first be implemented for a select group of residential customers and then eventually expand to other customers and commercial customers. By eliminating the use of monthly aggregates and using fine-grain hourly and daily detailed-level data, the model aimed to identify detailed patterns in consumption so that forecasts were more accurate when it came to consumers and timely in planning and setting rates. The functional requirements outlined in the forecasting model were intended to ensure the entire system will directly support Lakeland Electric's operational objectives. It would also be important for the system to handle both hourly and daily data so it could

accurately predict energy used, peak demand, and revenue. Using hourly and daily would be necessary to account for short-term fluctuations and customer usage in a way that should not be plausible using monthly aggregates. It was necessary for the model also to include historic weather and demographic data to represent the significant indicators of electricity consumption in the Lakeland area. In addition, a further requirement was to be able to implement the utility's residential rate structures (RS and RSX-1) so that forecasts could account for price-related influences on load behavior. Lastly, the model must generate outputs that are clear enough for staff to summarize insights, identify patterns, and make reasonable run planning or rate-related alternatives. In addition to functional requirements, a number of non-functional requirements were also considered to ensure that the forecasting model would meet the future operational needs of Lakeland Electric. For example, reliability was imperative, as the model needed to generate reliable forecasts on a consistent and repetitive basis, even during peak demand and/or extreme weather events. Scalability was also recognized and expressed as the system should, at some point in the future, be redeveloped to support future customer classes in addition to residential customers. As an operational consideration to ensure the model would be applicable, the model was required to be maintainable so that it could be updated with the most up-to-date information available, considering that, like rate structures and operational goals, data will also regularly change. Performance was also one of the higher priorities, as the model has to produce forecasts quick enough to support near real-time planning and operational decisions. Moreover, transparency and interpretability were paramount; staff needed to make sense of and have confidence in the model's predictions, especially when needing to make a case to stakeholders or regulators. Finally, the model needed to be robust despite the noise in customer usage behavior and weather variability and able to be used and remain stable even when input data are inconsistent and/or incomplete. These non-functional requirements were instrumental in developing a forecasting tool that was both technically sound and consistent with both short- and long-term practical and strategic objectives of Lakeland Electric. The development of the forecasting model was subject to several fundamental constraints that influenced the project's community and design. One significant constraint was the data's limited scope and clarity. The dataset that was provided only consisted of data from approximately five residential buildings and was thus only small in size but did not have enough diversity in customer types, as the set was limited to residential, where no commercial or real-time information was included. These issues made it difficult to understand the model's generalizability and applicability to the whole Lakeland Electric grid, and the economic and demographic data—particularly income and population—were often ambiguous or very poorly documented, which raised frequent questions on what they meant and if they were usable, adding to the complexity and making it difficult to incorporate these variables in any meaningful way. The team could only work with data Lakeland Electric provided directly; they did not have access to outside data or supplemental sources that broadened the analysis. In addition to the data-based complications, there were also time constraints on the project—the process needed to be efficient, focused on creating a residential load model to use and analyze rather than using

the time on real-time or larger customer forecasting. Despite all these limitations, the team aimed to keep it as modular as possible to ensure the model would be able to scale or enhance in future iterations with available data.

3. Concept Generation and Alternatives Analysis

To help inform the development of our load forecasting model, we completed a comprehensive test and product review, which draws from a combination of both academic and industry sources. The analysis assessed around 25 peer-reviewed research papers to determine optimal elasticity ranges for the key variables influencing residential electric load, including electricity price, GDP, population, wind speed, heating degree days (HDD), and cooling degree days (CDD). Those elasticity values provided necessary data to build the final logarithmic load prediction equation that would help to model how demand may change to adjustments in these variables. Also, we reviewed the modeling practices commonly used in established forecasting models, including Itron's SAE econometric model, which is a leading model for system-level forecasts of residential energy usage, and statistical models, including linear, SARIMAX, and Holt-Winters exponential smoothing and related statistical models, which provided both the basis for expected baseline performance and guidance for model refinement. By drawing evidence from both academic literature and modelling techniques applied in industry, we were able to ensure the forecasting models were developed on a base of evidence and research-informed evidence. Our goal was to test many different forecasting strategies briefly. We established a brainstormed concept in commonality of literature and gained practical experience and examination of multiple model concepts along the way in a relatively fast and repetitive process to validate the assumptions and performance described above. Initially, we envisioned a research-centric baseline linear regression model, which we built using elasticities extracted from about 25 peer-reviewed articles. We established a theoretical basis and consistent way of interpreting key features such as HDD/CDD, wind speed, GDP, price, and population in logarithmic format to retain our elasticity-based interpretation. After this, we continued with an OLS-style model specific to the Lakeland dataset, allowing us to compare coefficients we derived from research with coefficients we learned solely from local data. From there, we explored sensitivity analysis, where we varied the elasticity of each feature across the high and low ends of each feature's elasticity based on the documented high and low bounds while averaging all other features. We were able to identify how elastic our load predictions were to each input specification while also refining how we interpret the model working under different assumptions. We also started incorporating time-series forecasting models, such as SARIMAX and Holt-Winters. SARIMAX, which was ultimately selected because of its ability to address seasonality and to handle external factors, was a truly arduous task to get operational, particularly because it required almost a full 24 hours of computational time on a very large dataset to run a forecast of one month, using a SARIMAX model with (1, 0, 1) as the parameters and a seasonal component with (1, 0, 1, 24). We couldn't get it to run for longer horizons, suggesting both our computational restrictions as

well as the spatial considerations of implementing a real-world forecast. Holt-Winters, by contrast, ran easily as a simple benchmark model based on historical load only. Finally, we also tried a random forest, which makes an ensemble of decision trees using bootstrapped samples of the data. Here we used 100 estimators, a maximum depth of 20, and a minimum split of 5, which trained on 80% of the data and tested on the remaining 20%. The residuals from this model were relatively normally distributed around zero, although we had a slight bias toward underprediction. The random forest—importantly—was quite robust; however, I did not like that there was a degree of decentralized interpretability compared to the linear models, and it limited what we could communicate to stakeholders in terms of relevant actions. The generation of concepts involved both interpretable and black-box models, made note of both strengths and weaknesses, and we were able to explore promising paths for continuing development, all dependent on the unique challenges of the Lakeland data set. The primary selection criteria for evaluating model performance was Root Mean Square Error (RMSE). RMSE was chosen as the metric of choice since it uniformly describes the average size of forecasting errors and emphasizes larger errors to gauge reliability in energy load forecasting applications. Other factors such as interpretability, scalability, and ease of implementation were also important to evaluate, but RMSE was primarily our quantitative performance metric, since it provided a uniform basis to evaluate models by forecasting horizon and pricing structure. Alongside RMSE, we relied on a weighted decision matrix to ensure those key practical dimensions were balanced. Accuracy was weighted most heavily (40%) given the significance of this aspect, followed by interpretability (25%), efficiency (15%), stability (10%), and flexibility (10%). This system allowed us to assess the strengths and weaknesses across each model application beyond just accuracy. Trade-offs were unavoidable. For example, while random forests returned good RMSE metrics for longer forecast horizons, interpretability and stability were both coming up short (both are important for model transparency and to sustain for subsequent forecast periods). Conversely, SARIMAX excelled in terms of accuracy but showed poor computational efficiency and consequently scalability, particularly for multi-year forecasts. Holt-Winters had good implementation efficiency and ease of use but experienced significantly unstable results in the long term. OLS (fitted) was much more interpretable and efficient, making it an easy go-to for explainability and end-user utility; however, its predictive ability could not match upcoming models. In the end, the research model was the best middle ground. It was the only model to have high accuracy for all horizons and scored a balance on interpretability, efficiency, and stability. It also had a weighted decision matrix score of 89 out of 100, which made it easily the best option. By combining RMSE and our weighted evaluation framework, we were able to support the selection of the research model as the singular concept. It provided a better trade-off between predictive performance and practical deployment needs while still in line with our overall project goals. To provide a view of the entire process that occurred in developing the forecasting models, we produced a flowchart illustrating the entirety of the design process. The flowchart highlighted distinct phases, including the literature review, the generation of concepts, the development of models, the evaluation of performance, and the selection of

the final model. It was a visual plan that helped to structure our iterative process while communicating to an audience how the project was aligned with project goals and objectives. Appendix 6G presents the entire flowchart. The electric load forecasting system put in place by Lakeland Electric is built upon six main functional components: data preparation, model selection, quantitative evaluation, weighted decision matrix, trade-off analysis, and final selection. The process begins with data preparation, and we remove issues with the datasets of electric load and the explanatory variables, which are defined by missing values and typically not being aligned. After we clean the datasets, the data will serve as the basis for a strong model comparison and selection. In the model selection step, a variety of forecasting models are compiled: ordinary least squares (OLS), a research model that is based on elasticity assumptions for the industry, random forest, SARIMAX with exogenous regressors, and Holt-Winters exponential smoothing. Each of these models was put through the Quantitative Evaluation, where each model's performance was reviewed based on how well the forecasting was done by the best Root Mean Square Error (RMSE) over three different horizons (short-term (1 month), medium-term (1 year), and long-term (10 years)). A weighted decision matrix was also used to score models across 5 dimensions: accuracy (40%), interpretability (25%), computational efficiency (15%), long-term stability (10%), and flexibility to pricing schemes (10%). The weighted scores will support a thorough trade-off analysis that allows for real-world attributes associated with model implementation to be taken into consideration, such as stakeholder transparency, computational complexity, and being useful across a reasonable timeline. Lastly, the final selection combines empirical evaluation with a weighted decision matrix, which will rely on the final selection process to consider available alternatives and select the most appropriate forecasting model. The presented analysis suggests that the research model was the model that would provide the best trade-off between accuracy, interpretability, and usability for policy and pricing analysis in Lakeland Electric's context.

4. Applicable Codes and Standards

While there were no formal safety or regulatory codes governing electric utility forecasting, we were guided by the best practices of the electric utility industry. The use of Itron software, consistent with industry standards, informs our comparison of forecasting tools. In developing models independently, we followed principles of methodological rigor, transparency, and reproducibility that are consistent with university and industry experiences, and, importantly, ensured that we did not overfit our models and relied on rolling window forecasts and bias corrections whenever ambiguity persisted. Although this project was not subject to ANSI or IEEE standards, we took into consideration the best practices of data science concerning assumptions, fairness, and various evaluative criteria in utilizing quantitative and exploratory multi-criteria analysis.

5. Prototyping and Design Details

We developed proof of concept for this project through a full-scale Excel-based forecasting model that served as the inception of our modeling. The original model served as a "catch-all" Excel file that combined the academic literature, industry research, and elasticities into a working calculator for our analysis. This model configuration was based on forecasting equations used in the academic literature concerning energy demand forecasting. As a result, we used explanatory variables, such as electricity price, GDP, population, Heating Degree Days (HDD), Cooling Degree Days (CDD), and wind speed. Elasticity ranges for each variable were extracted from over 25 peer-reviewed research articles, which were coded into the Excel model and were used to simulate expected changes in electric load. This early model served as a quantitative demonstration of project viability; it showed a usable output based on actual data. This Excel prototype was a significant advancement from our original modeling plan, which focused on a more generic set of predictors such as historical load, short-term usage trends, mixed weather categories, etc., and a general catch-all for everything else, like day-of-week or economic activity. Although a good starting point, the early model was not economically interpretable and did not utilize elasticity values to show the influence of demand driver values such as price or income. The early model also excluded a number of key explanatory variables, which would later become critical to the performance of the model. As a result, while the original model was helpful to conceptualize the structural understanding, it provided no realistic simulation of policy-relevant scenarios or offered sensitivity analysis. The shift to the elasticity-based Excel model could fill in the gaps, providing a clearer empirical representation of load behavior. In addition to simulating the expected finite direction and size of load responses to external drivers, researchers could interrogate the expectations of the model's logic using ranges that drew on the existing academic knowledge in the relevant literature. Each data source, assumption, and citation was also tracked in the Excel file with sufficient detail and specificity to provide a high degree of transparency and traceability. This proof of concept also provided the foundation for every modelling phase that followed and played an important role in linking theoretical models with the real world. The focus of this project is on building a forecasting model, so the focus of design implementation is effectively "building" and "improving" that model by going through iterations and learning. The starting point for the core design was an Excel tool that was built as proof of concept for the forecasting process. The tool was based on over 25 research papers on energy demand forecasting and was built with variables like electricity price, GDP, population, heating degree days (HDD), cooling degree days (CDD), and wind speed. These input variables were used in the model to estimate electric load based on reasonable elasticity assumptions and in a way that was most useful for forecasting purposes, but also to provide a foundation for an informed and data-driven forecast. The design implementation in this instance was not based on features of typical design implementation, such as CAD drawings or wiring diagrams, but on design implementation of the model itself. The Excel file was a hub for all variables, data, and assumptions related to the project and was structured in such a way that enabled ease of instability of variables and for testing of hypotheses. The model could be easily adapted by the user changing input data to different values and understanding the

timetable impact of changing the assumption of a variable. The model was adaptable for different time horizons and conditions because it didn't enforce input data change on a weekly, monthly, or yearly basis. The result of this was a flexible model that designers could use in their forecasting role while comfortably experimenting in the model and seeing in real time how the changes made the forecast respond, and it gave them choice. During the development phase, the model transitioned from a basic proof of concept into a more advanced set of forecasting tools. The original proof of concept showed that the forecasting equation was able to produce outputs that were realistic and reasonable. Afterward, other forecasting methods (SARIMAX, Holt-Winters, and Random Forest) were tested, and their performance was evaluated using the Root Mean Square Error (RMSE) model performance metric across short, medium, and long-term forecasting horizons. The design implementation was guided by a clear structure for model evaluation. A weighted decision matrix with trade-off analysis was employed to guide the selection of the chosen model, ensuring that the design was both functional and met the project definition of accuracy, interpretability, and computational efficacy. Therefore, although physical design elements were absent, the design implementation of this forecasting model consisted of an iterative approach of improving the model's structure, predicting accuracy, and practical fit with Lakeland Electric's operational needs. The final outcome was an advanced, transparent, effective forecasting tool that would be easily incorporated into real-world applications. There were two main phases of the development of the forecasting model: the Alpha prototype, which was developed in Excel, and the Beta prototype, which was developed across a range of modeling platforms. The Alpha prototype served as a single proof-of-concept calculator. It was entirely created in Excel and implemented elasticity-based forecasting equations from over 25 academic and industry articles. The model was focused on manually inputting elasticity ranges for important variables, such as electricity price, GDP, population, HDD, CDD, and wind speed, while also including a comprehensive citation structure and a clearly articulated assumptions section, thereby allowing for the early verification of structure and feasibility and demonstrating a logical behavior of the model with respect to changes in inputs. At that time, it was particularly useful for confirming that the forecasting equation could produce reasonable outputs based on foundational empirical research similar to most forecasting methods. The Beta prototype introduced more formalized modelling methods using OLS regression, SARIMAX with external variables, Holt-Winters probability forecasting, and Random Forest, based on standardized datasets, and evaluated performance using RMSE. Nonetheless, while it expanded to incorporate more automation and scale, the beta model still retained some simplifications. For instance, there were no dummy variables to account for seasonality or categorical effects (certain months or changes in policy). Instead, it relied on earlier and, therefore, less refined estimates of elasticities that had not been optimized and, therefore, narrowed yet. The limitations led to inconsistencies, particularly around long-term forecasting and price sensitivity simulations. Team feedback and results from comparative testing provided key revisions. The ranges of elasticities were narrowed based on additional synthesis of the literature and professor feedback, and the assessment of the importance of interpretability

and feasibility of deployment was revisited. These reflections helped return to re-emphasize the research model as it provided a "clearer" trade-off between precision and usable applicability for policy and pricing decision-making. The beta model not only started to discuss more advanced modeling techniques, but it also included important structural features, for example, in its support for both static and dynamic pricing processes. These pricing configurations set up the possibility of future probing of sensitivity analysis and rolling window forecasting, which together can evaluate any model's ability to change as market conditions change and evaluate an appropriate time horizon for reliability. While not fully tested in this beta phase, they were built into the model we created and will be discussed further in the "Testing and Evaluation" section. Overall, these components represented a shift to a policy-oriented forecast, not strictly accuracy, but continuous usability in existing utility settings in real-time.

6. Testing and Evaluation

Instead of concentrating upon hardware components like flow rate, sensor accuracy, and Bluetooth reliability, the project was largely focused on model validation using advanced analytical means. Two methods were used most prominently in robust evaluation of forecasting robustness, performance reliability, and actual real-world behavior: sensitivity analysis and rolling windows analysis. Sensitivity analysis was a key diagnostic method to make assessments on quantified changes in the key explanatory variables (heating degree days (HDD), cooling degree days (CDD), income, and population) that resulted in changing predicted load demand. This method allowed a "what if" type of testing to be performed and, using plausible future scenarios and/or policy options, assessed the responsiveness of the models. Most importantly, however, the analysis showed the significance of using Lakeland-specific elasticity values that were vital in capturing customer responsiveness rather than typical national or global elasticities. This analysis also used research-based elasticities (average elasticities) and zero elasticity values to compare them with the majority of the observed outputs and would then measure the effect of external variables on load behavior. This comparative framework was instrumental in validating not just forecasting accuracy but also the behavioral realism incorporated in the models. Rolling windows analysis was another cornerstone of model validation. It was used to evaluate how forecast accuracy changed over the historical training data by iteratively changing the training data window and forecasting to the future. This method simulated the actual world of forecasting and stressed the models' ability to generalize in time. Rolling windows analysis dealt with HRL data, with static and dynamic pricing scenarios represented in the data windows themselves. This format allowed the models to infer not just long-term trends in demand at prices but short-term responses to changes in prices. This same concept was used for hourly and daily bias corrections aimed at better aligning what the model predicted compared to what occurred, and to a lesser degree, the baseline model's performance was improved, and the systematic nature of the prediction error was reduced. The rolling structure successfully incorporated seasonality (including 24-hour load cycles)

and long-term drift in a principled, repeatable way. The rolling windows model provided a pragmatic context for developing a sense of stability in the models and their flexibility in responding to dynamic conditions. By updating the training datasets each time new information emerged, the models were challenged to forecast at multiple time frames and under different economic environments. This gave a profound sense of the models' performance under various conditions, whether it was a year with extreme weather event loads or a year of economic growth. With the implications of both static and dynamic pricing under each rolling window, this also enabled the testing of some policy outcomes, especially the role of pricing structures on load demand. The models with dynamic pricing assessed the models' responsiveness to price changes, whereas the static pricing models acted as benchmarks on the performance of the models given a more traditional pricing system. In this way, rolling windows allowed for prediction validation, as well as asking whether they remained relevant and capable of flexibly addressing real-world problems, which is a critical dimension of insights into the extent to which future forecasting approaches may be enhanced.

7. Risk Assessment and Mitigation

One of the important early concerns was leakage of data and overfitting issues, as we were using machine learning-based models like Random Forest, which are complex and can memorize the noise in the training data, where generalization suffers. We were conscientious to apply regularization and filter the evidence through cross-validating over multiple data splits to assess some degree (we are cognizant of the limitation here) of stability in the models and identify models that predict similarly on both training and unseen data. Further, we had the advantage of conducting a sensitivity analysis through the real-world volatility, including vanishing equities in Florida, like hurricanes in October that caused outages and behavior deviation and trends for energy use. These phenomena provide realistic stress testing on the models. Additionally, we took into consideration observable seasonal patterns that led to behavioral reductions in AC use in the winter. These solutions helped create more robust models and build trust with their predictions.

8. Conclusions and Future Work

Through the use of time series and econometric models, this project enhanced Lakeland Electric's capacity to predict residential electricity demand; OLS regression proved to be the most successful. Demand is probably significantly impacted by pricing structure, which is especially noticeable in localized data results. OLS required less tuning and provided more parameter transparency than Random Forest. Additional behavioral insights were obtained by modeling elasticity at the granularity we selected. Rolling windows and cross-validation guaranteed model reliability and ought to be regarded as standard machine learning procedures. To improve temporal resolution, future work should incorporate additional customer segments and substitute hourly weather data for HDD/CDD. Adaptive forecasting and real-time updates will soon be possible with smart grid integration.

9. Individual Contributions

Team Member	Contribution
Josiah Jackson	40%
Nicholas Ricci	20%
Samuel Hesse	20%
Wenslo Dort	7%
Maria Valencia	10%
Jackson Marulis	2%
Jovan Ortiz	1%

Appendix 1: Full List of Customer Needs, Objectives, and Requirements

Customer Needs:

- Accurate load predictions to reliably meet growing electricity demand without service interruptions.
- Fair and manageable rate structures for Lakeland Electric's customers.
- Models that use Lakeland-specific data and reflect real usage behavior and local conditions.
- Ability to account for future growth, anticipate shortfalls, and support decisions on electricity purchases or rate adjustments.
- Initial focus on residential customers—the largest source of new demand.
- A scalable pipeline that allows expansion into other customer types (e.g., commercial, industrial).

Project Objectives:

- Develop a forecasting model using real customer data from Lakeland Electric.
- Prioritize interpretability and accuracy for utility planning and communication.
- Utilize hourly and daily data instead of monthly aggregates to improve model precision.
- Create a working model for a defined set of residential customers that can later be scaled.

Functional Requirements:

- Process hourly and daily data for energy usage, peak demand, and revenue.
- Incorporate historical weather, demographic, and economic variables.
- Implement residential rate structures (RS, RSX-1)
- Provide clear, interpretable outputs suitable for use by non-technical utility staff.

Appendix 2: Concept Sketches and Evaluation Matrices

Model	Accuracy	Interpretability	Efficiency	Stability	Flexibility	Weighted Score (100)
OLS (Fitted)	7	10	10	7	8	83
Research Model	9	9	9	9	8	89
Random Forest	4	5	6	2	6	45.5
SARIMAX	10	5	2	7	8	70.5
Holt-Winters	6	8	10	3	5	67
Weight	40%	25%	15%	10%	10%	100%

Figure 1: Weighted Decision Matrix

Load Model:

- $L(t,c) = \beta_0 + \beta_1 \cdot H(t,c) + \beta_2 \cdot U(t,c) + \beta_3 \cdot W(t) + \beta_4 \cdot X(t,c)$
- $L(t,c)$: Projected load at time t for building class c . (kWh)
- β_0 : Intercept (baseline load when all other factors are zero).
- $H(t,c)$: Historical average load at time t for building class c .
- $U(t,c)$: Current usage trend for building class c .
- $W(t)$: Weather data at time t (such as temperature, humidity, etc.).
- $X(t,c)$: Any additional features that may influence load, e.g:
 - Day of the week
 - Building occupancy levels
 - Economic activity
 - Special events
 - Etc.
- $\beta_1, \beta_2, \beta_3, \beta_4$: Coefficients for the features

Figure 2: Original Load Model

Appendix 3: Prototype Equation, Sensitivity Analysis, Rolling Windows Evaluation, and Performance Comparison Across Models

Linear Models (Research Based/Lakeland Specific OLS):

Equation

$$\begin{aligned}\ln(L_t) = & \beta^0 + \varepsilon_p \ln(P_t) + \varepsilon_{gdp} \ln(GDP_t) + \varepsilon_{pop} \ln(Pop_t) \\ & + D_{hdd} \cdot \varepsilon_{hdd} \ln(HDD_t) + D_{cdd} \cdot \varepsilon_{cdd} \ln(CDD_t) \\ & + D_{wind} \cdot \varepsilon_{wind} \ln(Wind_t) + \varepsilon_t\end{aligned}$$

Dummy Variables

$$D_{hdd} = 1 \text{ if } HDD_t > 0.5, \text{ else } 0$$

$$D_{cdd} = 1 \text{ if } CDD_t > 0.5, \text{ else } 0$$

$$D_{wind} = 1 \text{ if } Wind_t > 0.5, \text{ else } 0$$

Figure 3: OLS Model.

Variable	Elasticity Range	Dynamic Pricing	Static Pricing
Price	-0.188 to -0.005	-0.854	-0.0186
Income (GDP)	0.393 to 1.132	0.422	0.4342
Population	0.109 to 0.883	0.0515	0.0415
HDD	0.00 to 0.221	0.1801	0.1733
CDD	-0.66 to 0.335	0.2408	0.2351
Wind Speed	0.011 to 0.022	0.0707	0.0635

Figure 4: Elasticity Ranges from Research Papers [3]-[33] and Lakeland Specific Model.

Above represents our research-based linear regression (OLS) model for forecasting the electrical load for Lakeland Electrical. The variables included are load (L), Price (P), income (represented using GDP), Population (Pop), Heating/Cooling Degree Days (HDD/CDD), and Windspeed (Wind). Starting off the only building class/customer type being considered is Residential. Currently the only weather data Lakeland Electrical uses is HDD and CDD which

are explained further in Appendix D below. This provides a way for us to easily expand their existing models by incorporating additional weather data starting with Windspeed. We only obtained load data for 5 residential buildings located within Lakeland. Assumptions for pricing were made by assuming the entire dataset utilized static pricing for one analysis and then assuming the entire dataset was dynamic pricing for another. This allowed us to see how the data differed based on the pricing structure since we did not obtain information relating to what actual pricing structures were followed by these buildings.

Hours 1–12		Hours 13–24		Weekday	Correction
Hour	Correction	Hour	Correction		
1	-23.13	13	176.84	Monday	16.71
2	-46.80	14	203.13	Tuesday	23.12
3	-64.07	15	217.71	Wednesday	18.45
4	-74.13	16	221.27	Thursday	-8.09
5	-75.52	17	217.64	Friday	-10.46
6	-64.18	18	202.12	Saturday	-29.27
7	-47.33	19	174.13	Sunday	-8.62
8	-33.62	20	140.01		
9	0.76	21	110.56		
10	48.95	22	77.91		
11	96.37	23	40.49		
12	139.43	24	4.27		

Figures 5 and 6: Seasonal Adjustments for Research Based Model.

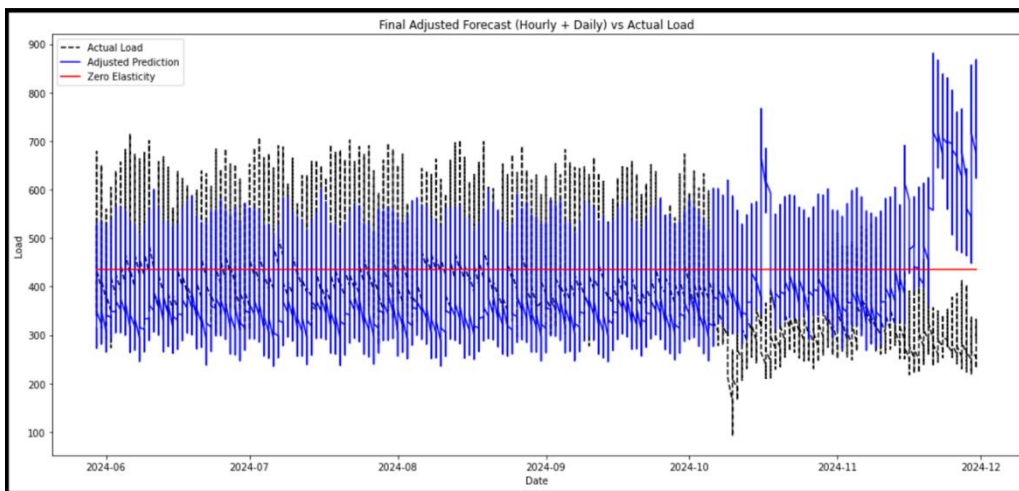


Figure 7: Research Based OLS Model

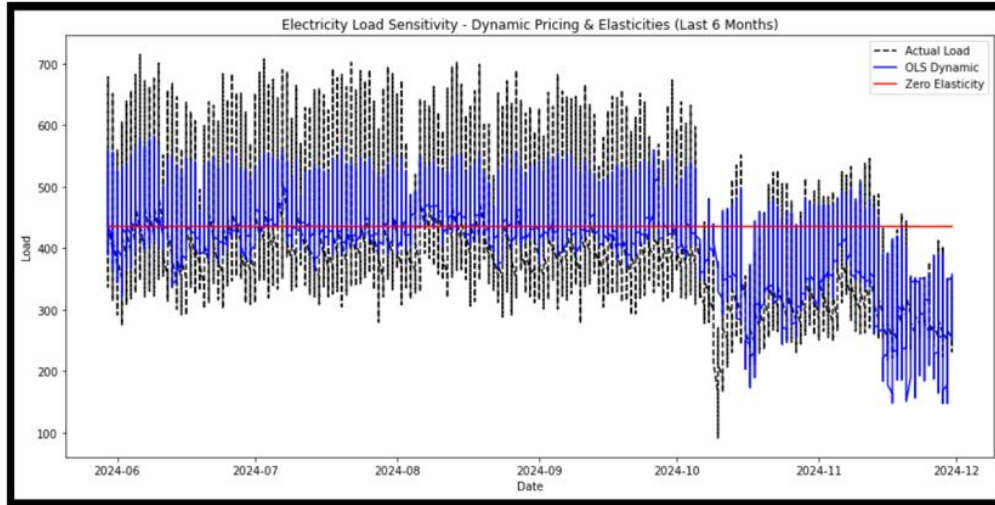


Figure 8: Lakeland Specific OLS Model

OLS Regression Results						
Dep. Variable:	log_load	R-squared:	0.547			
Model:	OLS	Adj. R-squared:	0.547			
Method:	Least Squares	F-statistic:	3.767e+04			
Date:	Fri, 28 Mar 2025	Prob (F-statistic):	0.00			
Time:	12:35:54	Log-Likelihood:	46467.			
No. Observations:	187019	AIC:	-9.292e+04			
Df Residuals:	187012	BIC:	-9.285e+04			
Df Model:	6					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	3.5880	0.062	58.337	0.000	3.467	3.709
log_price	-0.0085	0.017	-0.498	0.619	-0.042	0.025
log_gdp	0.4342	0.010	44.485	0.000	0.415	0.453
log_pop	0.0415	0.010	4.129	0.000	0.022	0.061
log_hdd	0.1733	0.001	228.310	0.000	0.172	0.175
log_cdd	0.2351	0.001	396.822	0.000	0.234	0.236
log_wind	0.0635	0.001	78.294	0.000	0.062	0.065
Omnibus:	0.667	Durbin-Watson:	0.238			
Prob(Omnibus):	0.717	Jarque-Bera (JB):	0.663			
Skew:	-0.004	Prob(JB):	0.718			

Figure 9: Static Pricing OLS Results

OLS Regression Results						
=====						
Dep. Variable:	log_load	R-squared:	0.552			
Model:	OLS	Adj. R-squared:	0.552			
Method:	Least Squares	F-statistic:	4.614e+04			
Date:	Fri, 28 Mar 2025	Prob (F-statistic):	0.00			
Time:	12:35:55	Log-Likelihood:	47523.			
No. Observations:	187019	AIC:	-9.503e+04			
Df Residuals:	187013	BIC:	-9.497e+04			
Df Model:	5					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

log_price	-0.8540	0.009	-96.758	0.000	-0.871	-0.837
log_gdp	0.4220	0.010	43.525	0.000	0.403	0.441
log_pop	0.0515	0.010	5.143	0.000	0.032	0.071
log_hdd	0.1801	0.001	234.906	0.000	0.179	0.182
log_cdd	0.2408	0.001	401.318	0.000	0.240	0.242
log_wind	0.0707	0.001	76.219	0.000	0.069	0.073
=====						
Omnibus:	16.219	Durbin-Watson:	0.231			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	16.227			
Skew:	-0.022	Prob(JB):	0.000300			
Kurtosis:	2.989	Cond. No.	317.			
=====						

Figure 10: Dynamic Pricing OLS Results

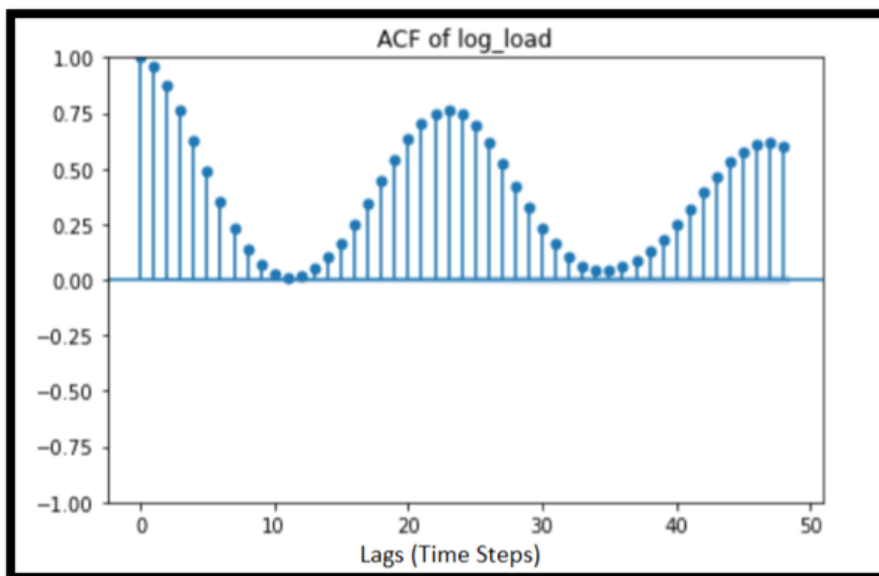


Figure 11: ACF for log(Load)

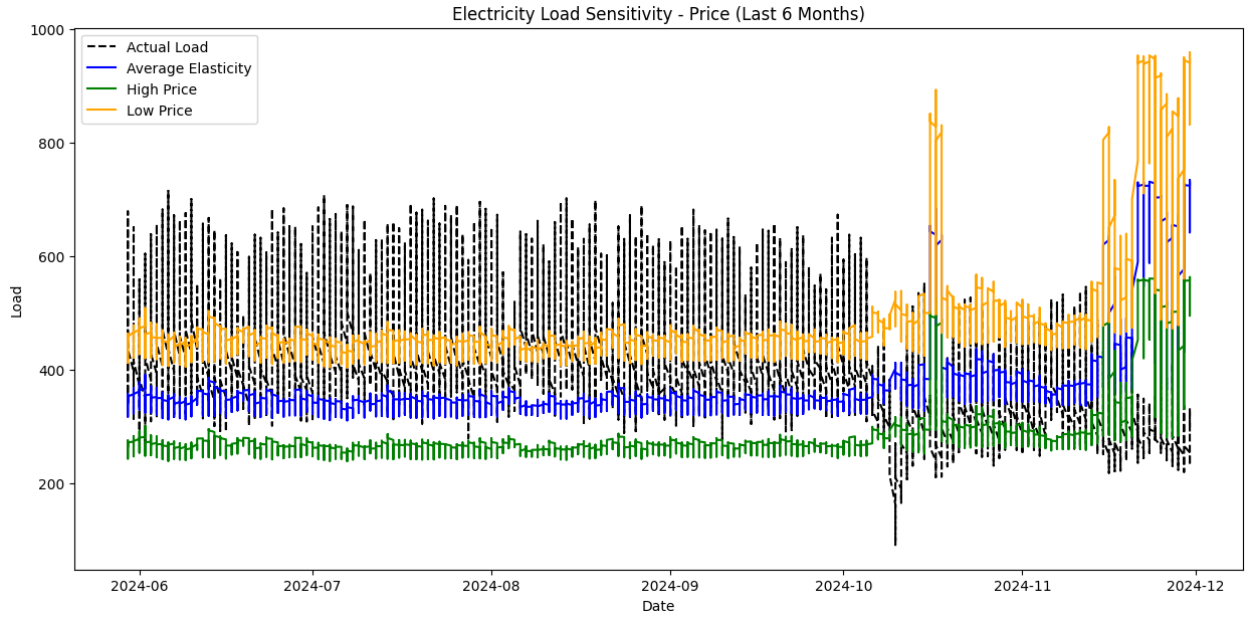


Figure 12: Sensitivity Analysis (Price)

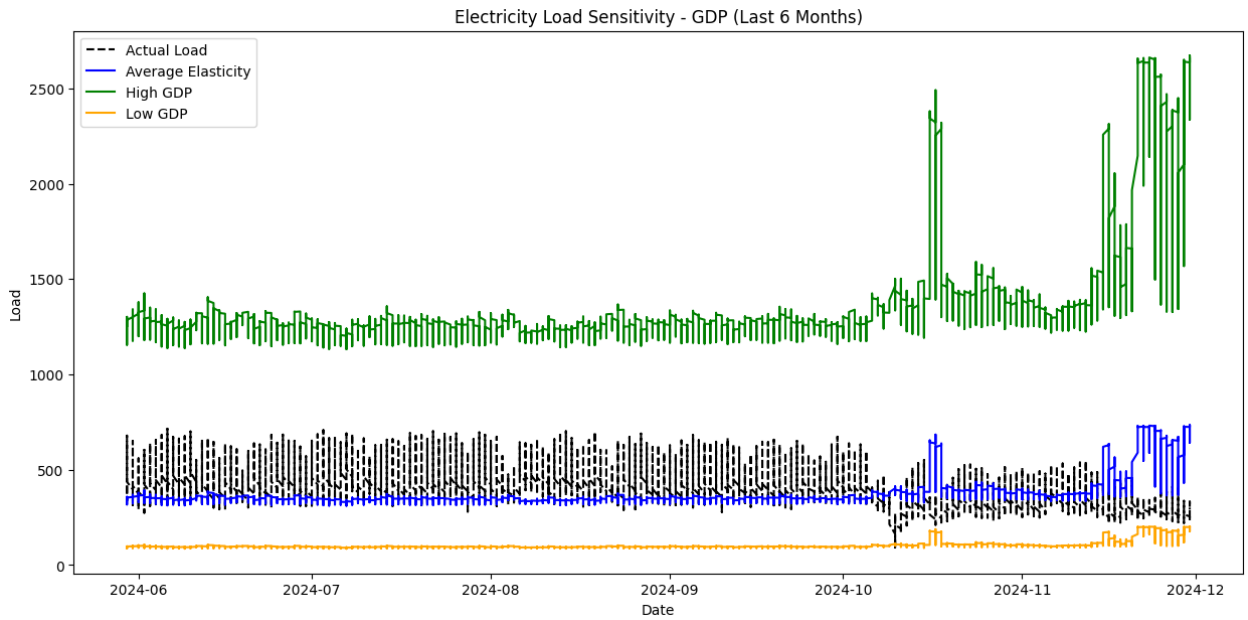


Figure 13: Sensitivity Analysis (GDP (Income))

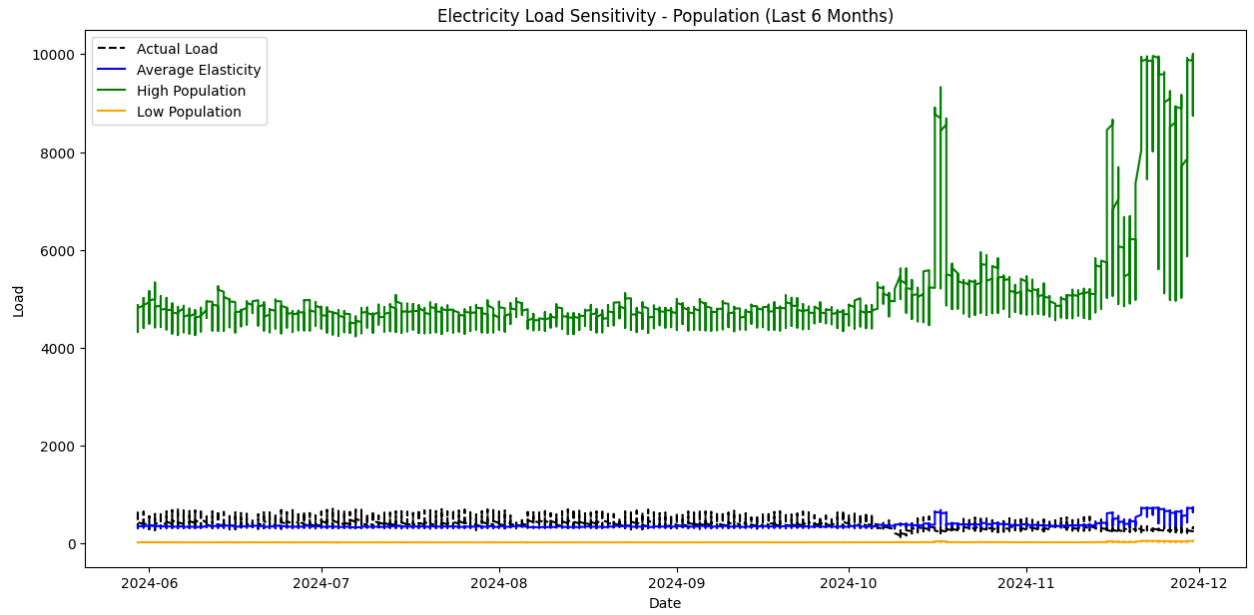


Figure 14: Sensitivity Analysis (Population)

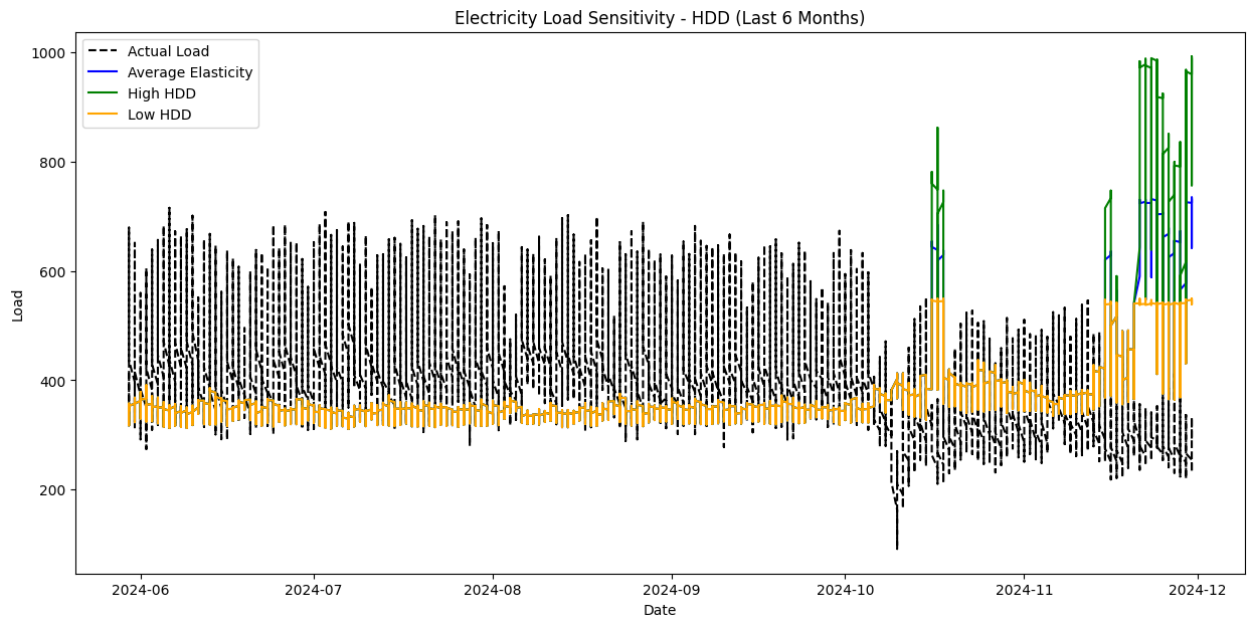


Figure 15: Sensitivity Analysis (HDD)

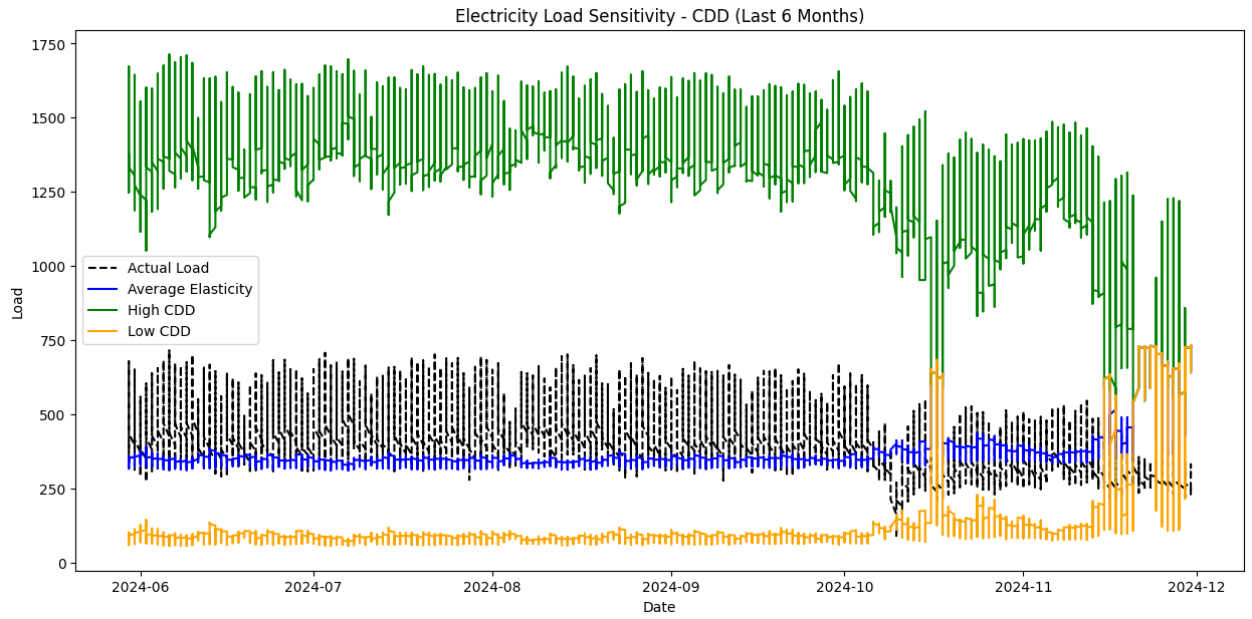


Figure 16: Sensitivity Analysis (CDD)

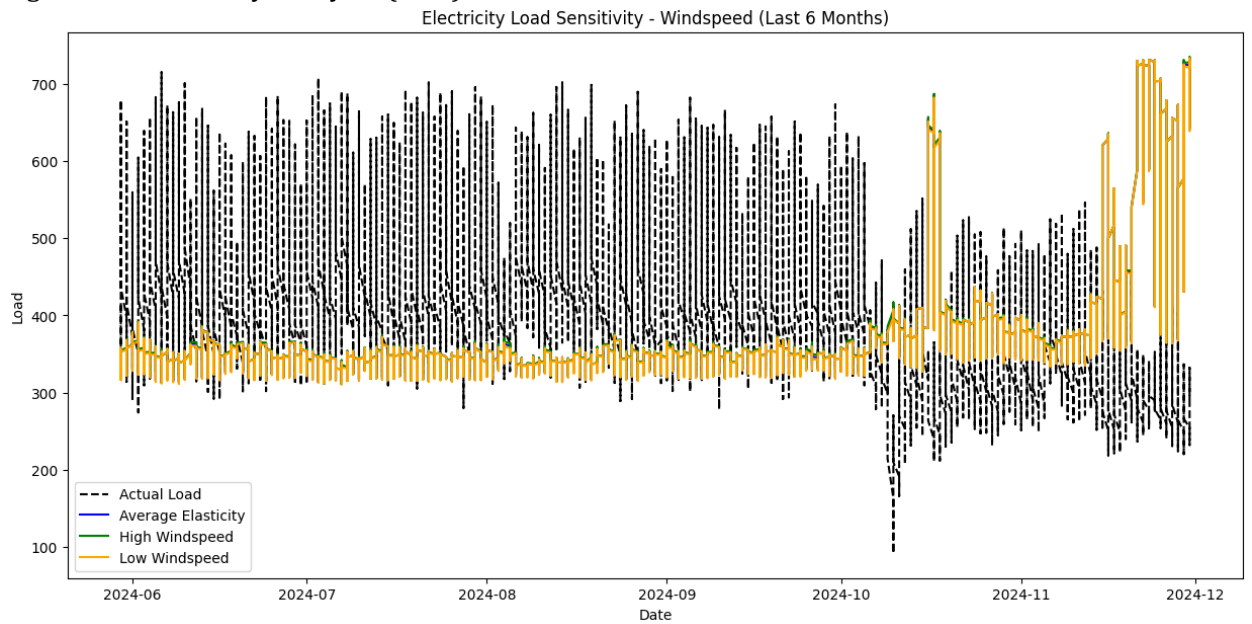


Figure 17: Sensitivity Analysis (Windspeed)

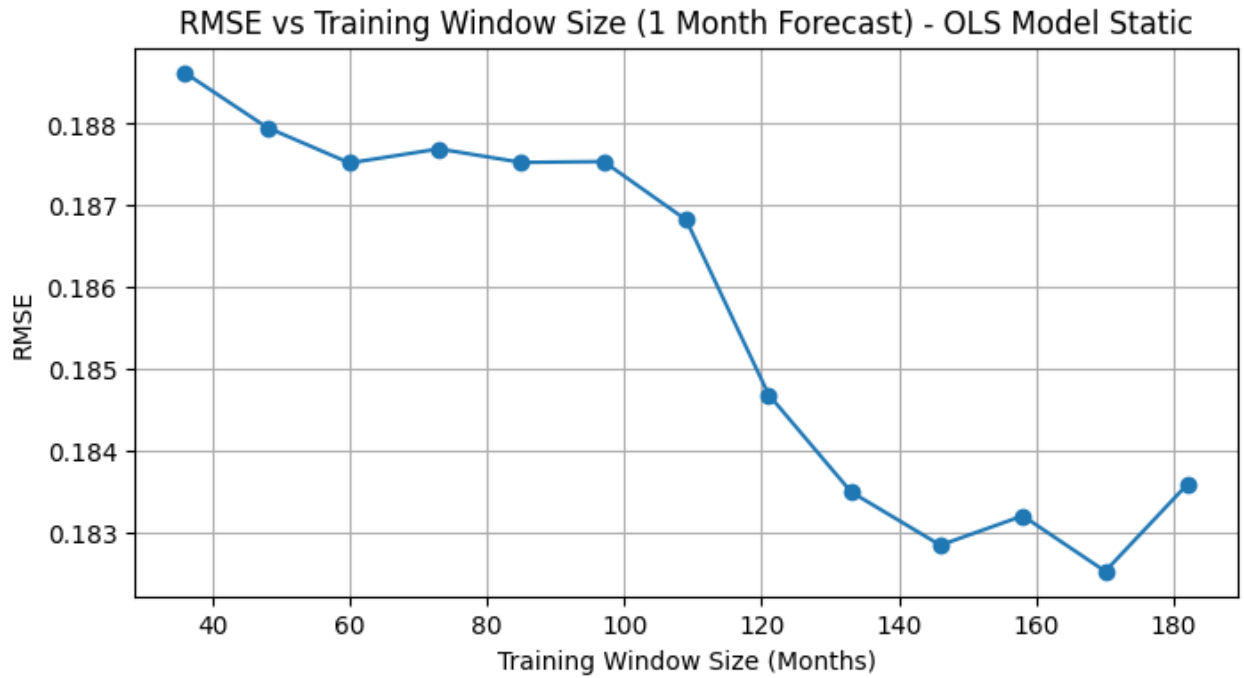


Figure 18: 1 Month Rolling Window – OLS Model (Static Pricing)

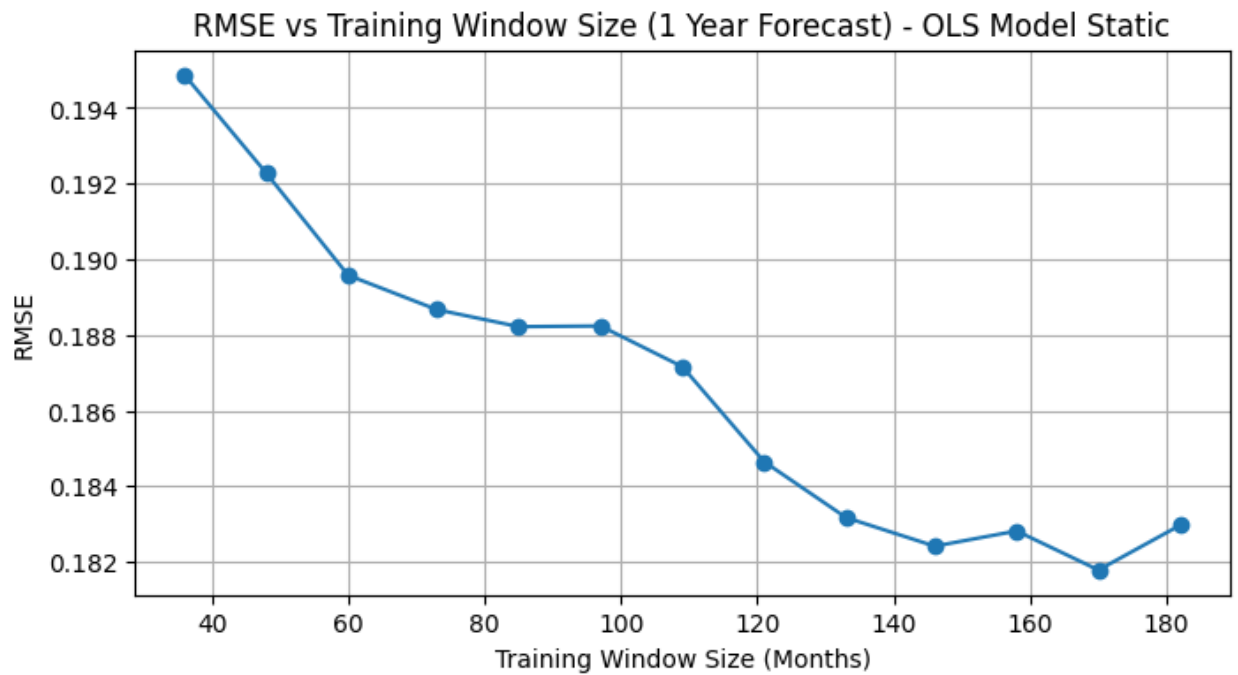


Figure 19: 1 Year Rolling Window – OLS Model (Static Pricing)

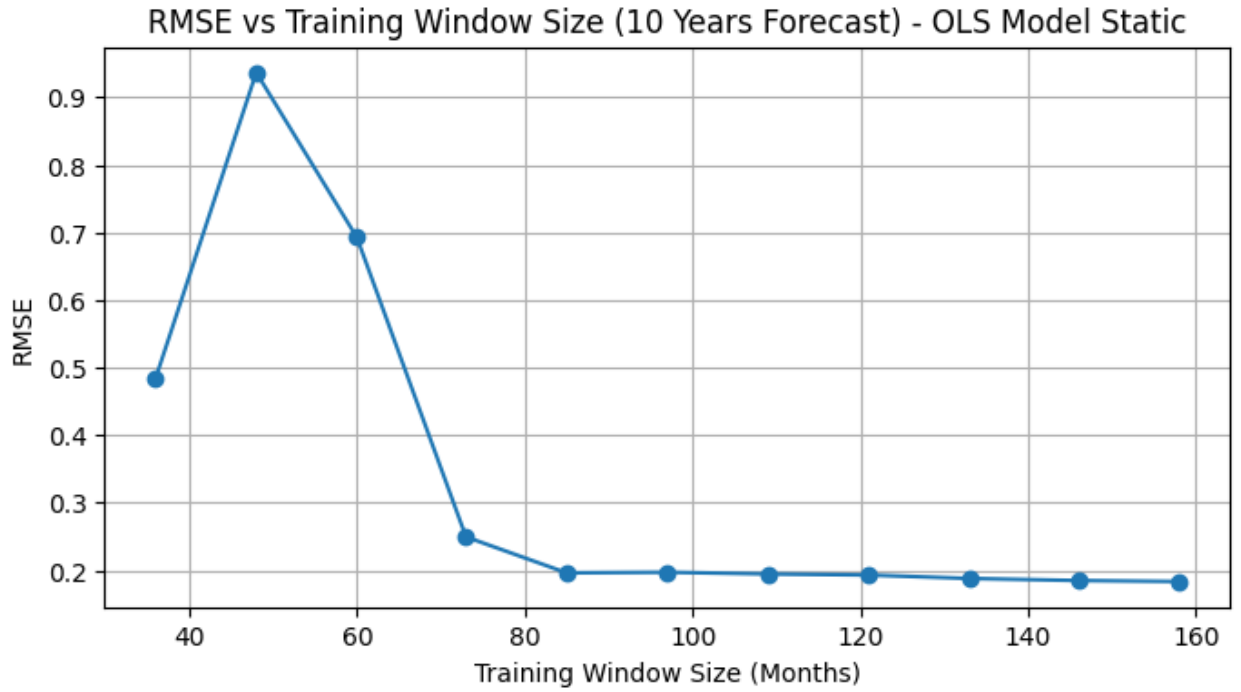


Figure 20: 10 Year Rolling Window – OLS Model (Static Pricing)

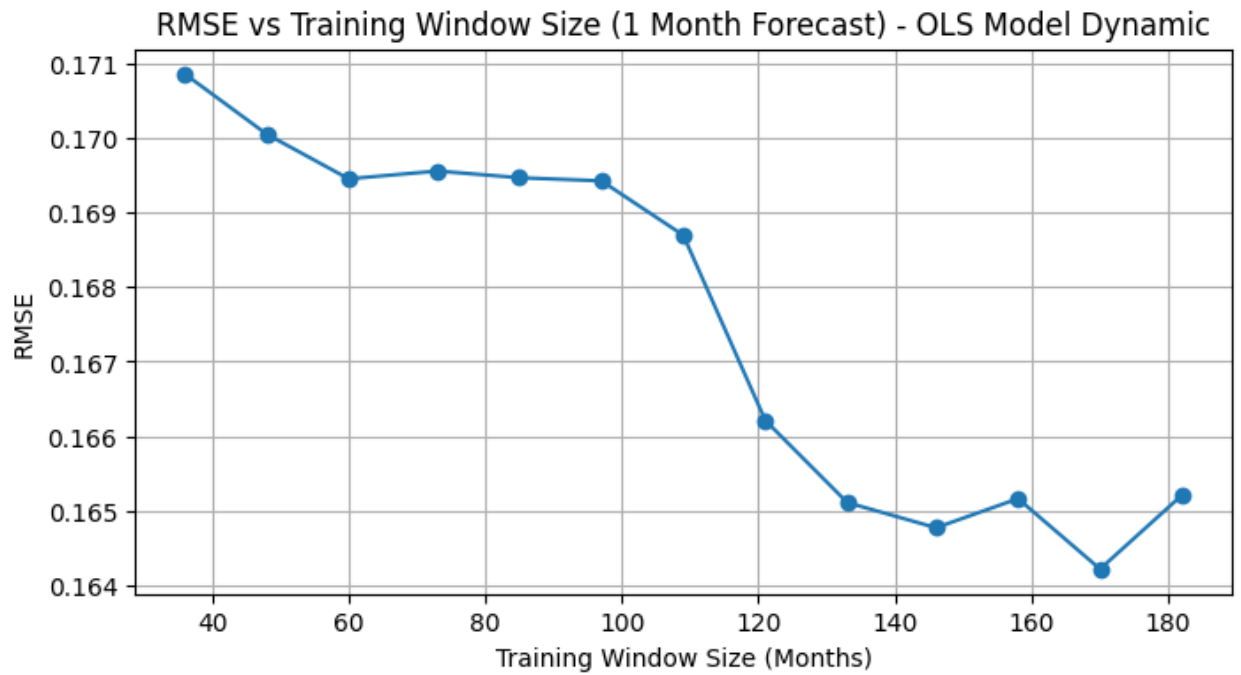


Figure 21: 1 Month Rolling Window – OLS Model (Dynamic Pricing)

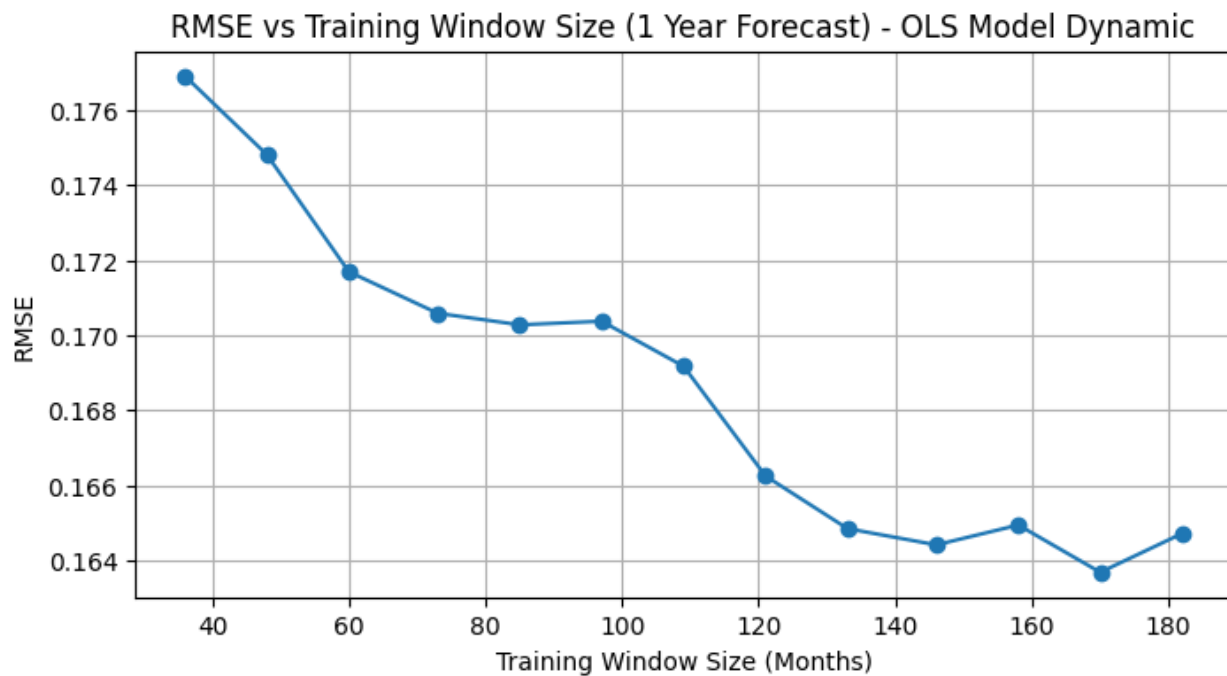


Figure 22: 1 Year Rolling Window – OLS Model (Dynamic Pricing)

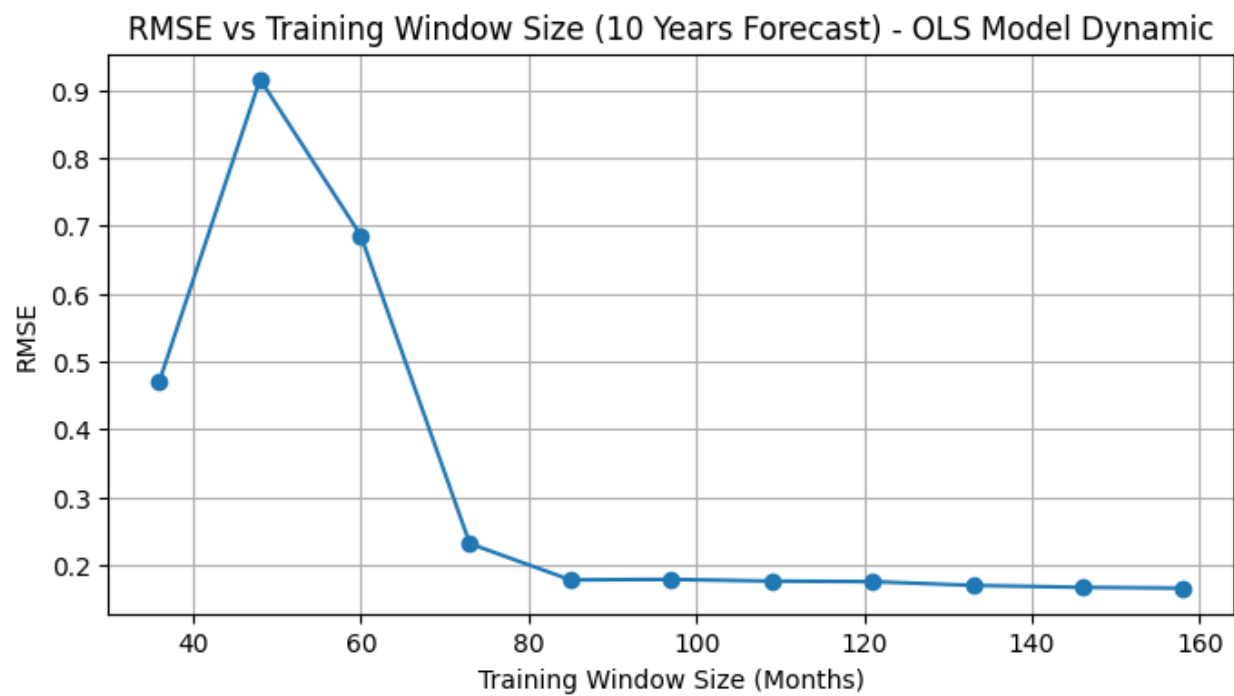


Figure 23: 10 Year Rolling Window – OLS Model (Dynamic Pricing)

Forecast Horizon	Model Type	Optimal Window Size (Months)	Optimal Window Size (Hours)	Forecast RMSE
1 Month	Static	170 months	122,640 hrs	0.1825
	Dynamic	170 months	122,640 hrs	0.1642
1 Year	Static	170 months	122,640 hrs	0.1818
	Dynamic	170 months	122,640 hrs	0.1637
10 Years	Static	158 months	113,880 hrs	0.1833
	Dynamic	158 months	113,880 hrs	0.1657

Figure 24: Table of Best Window Sizes for Rolling Windows

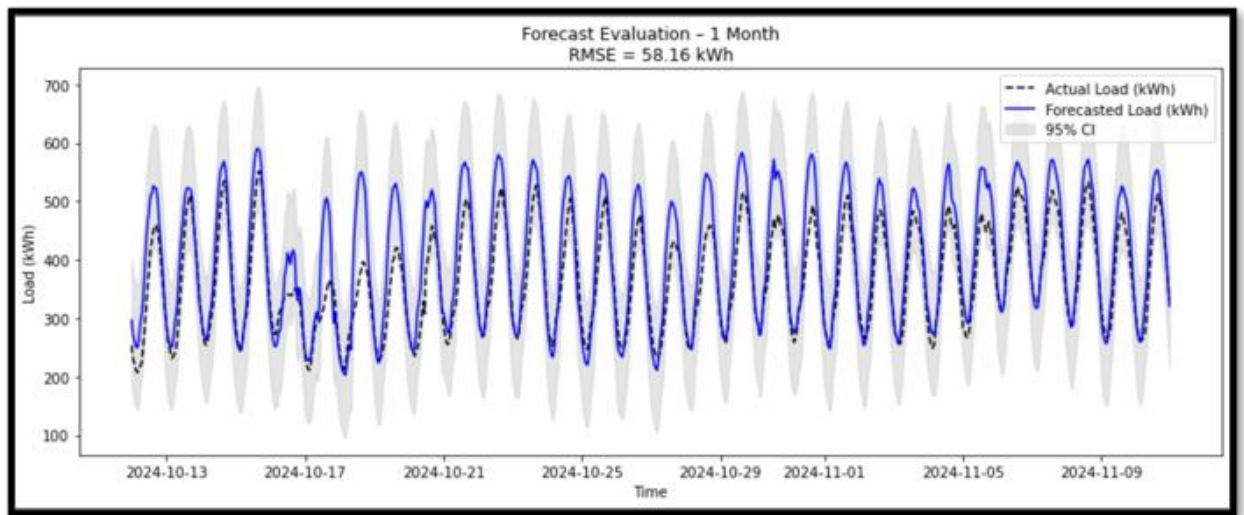


Figure 25: Forecast Evaluation - 1 Month - Static Pricing

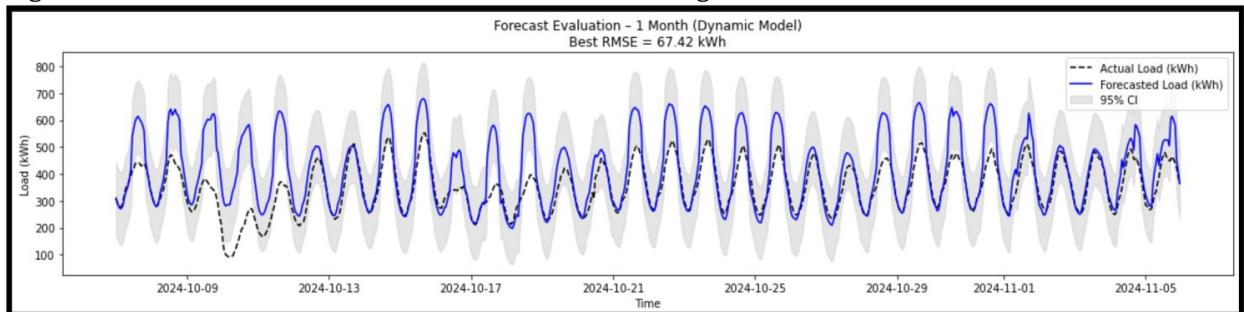


Figure 26: Forecast Evaluation - 1 Month - Dynamic Pricing

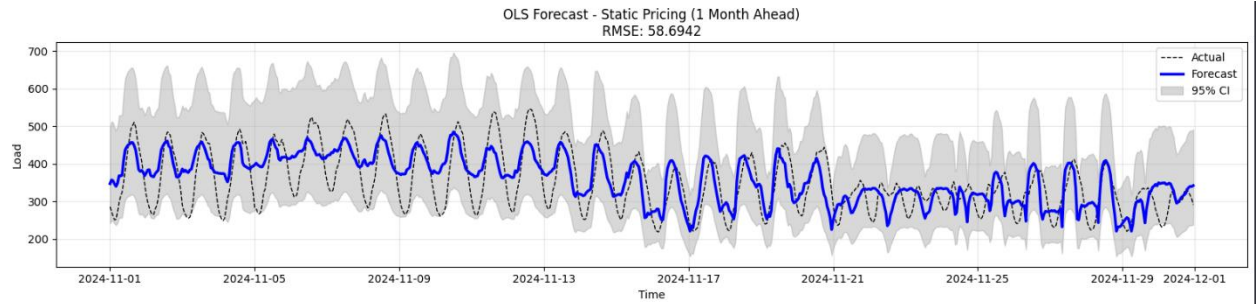


Figure 27: OLS 1 Month Forecast (Static)

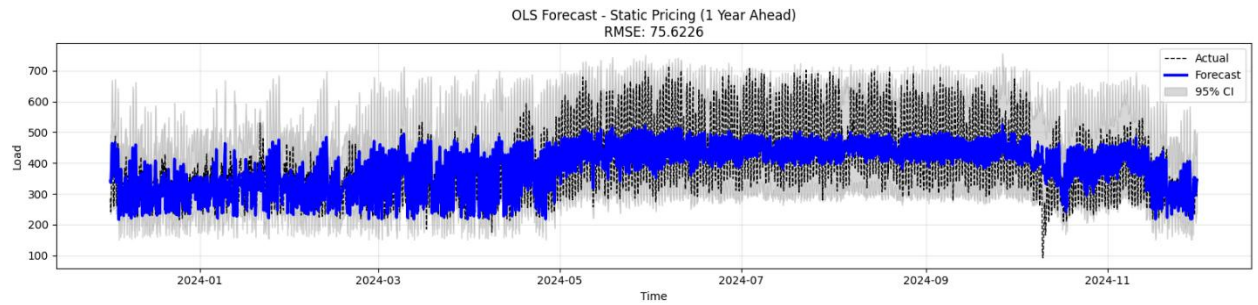


Figure 28: OLS 1 Year Forecast (Static)

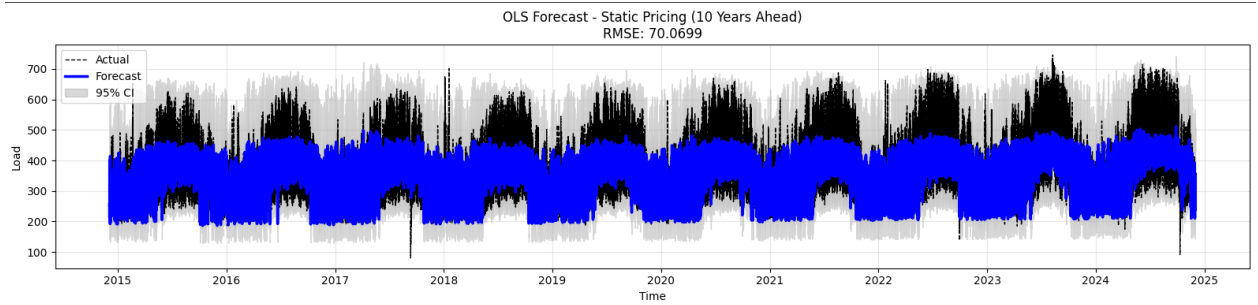


Figure 29: OLS 10 Year Forecast (Static)

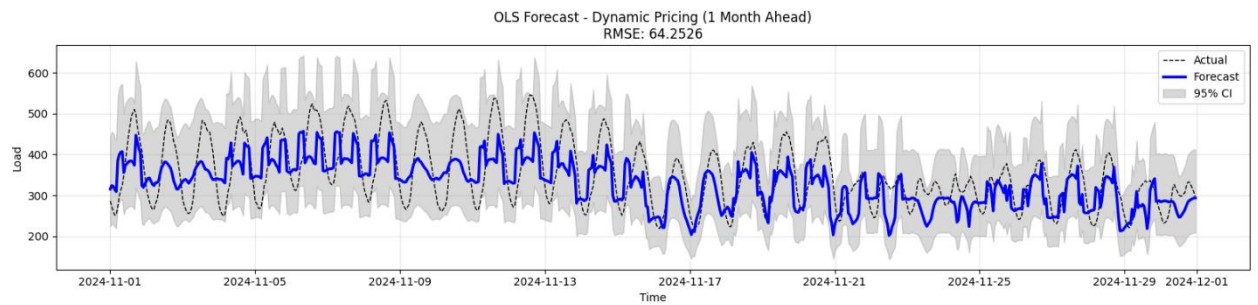


Figure 30: OLS 1 Month Forecast (Dynamic)

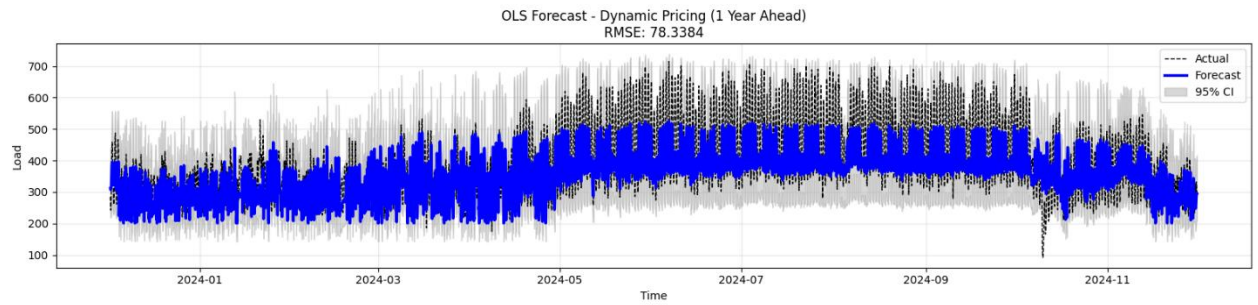


Figure 31: OLS 1 Year Forecast (Dynamic)

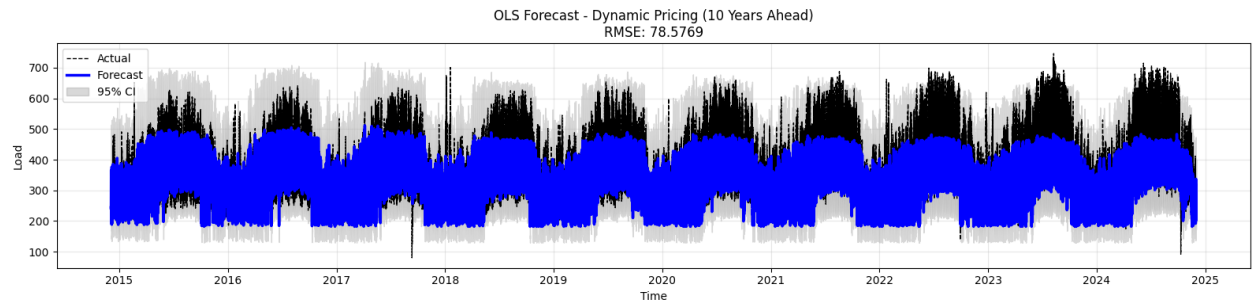


Figure 32: OLS 10 Year Forecast (Dynamic)

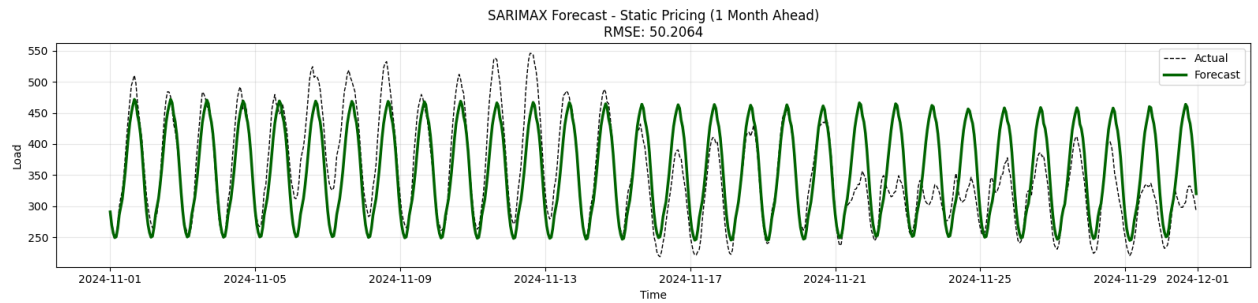


Figure 33: SARIMAX 1 Month Forecast (Static)

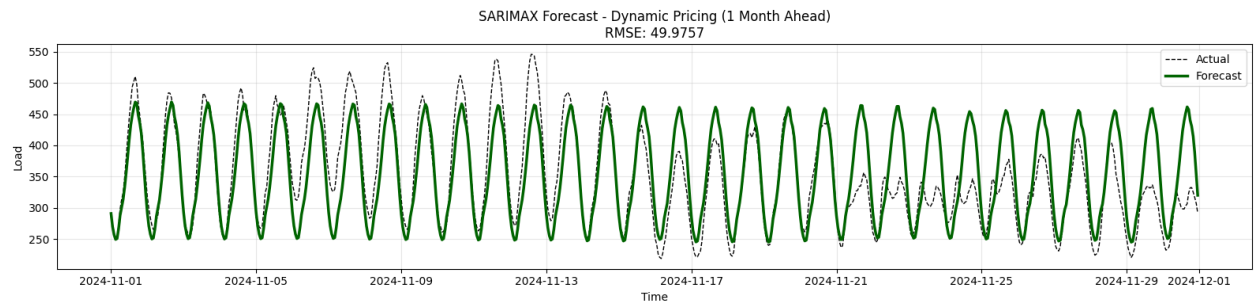


Figure 34: SARIMAX 1 Month Forecast (Dynamic)

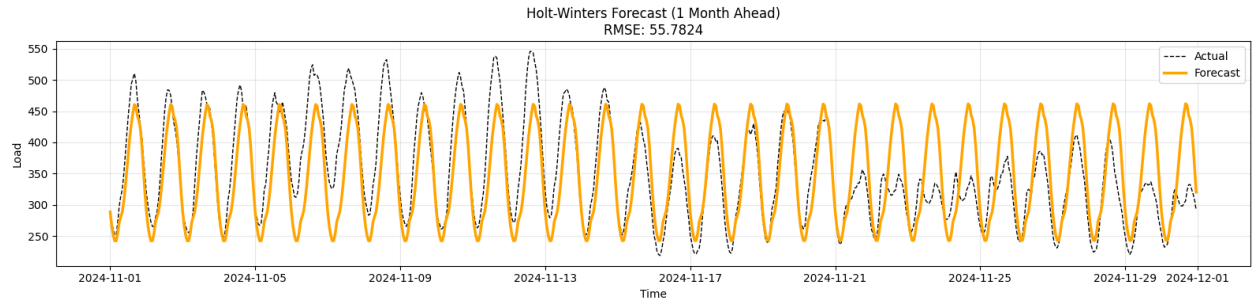


Figure 35: Holt-Winters 1 Month Forecast

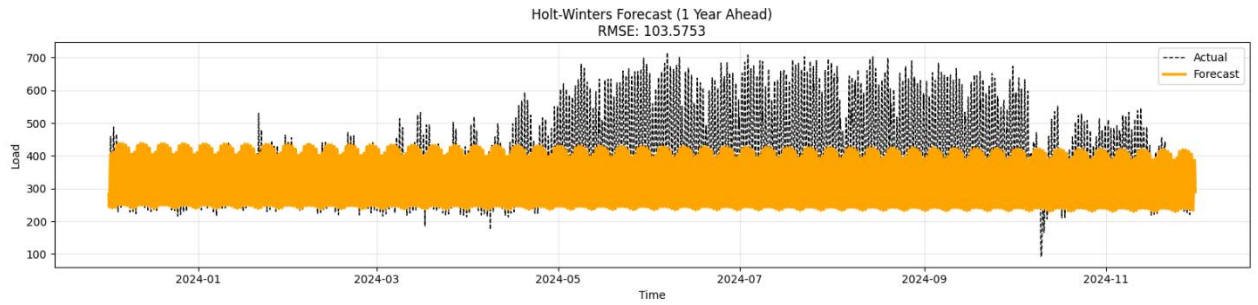


Figure 36: Holt-Winters 1 Year Forecast

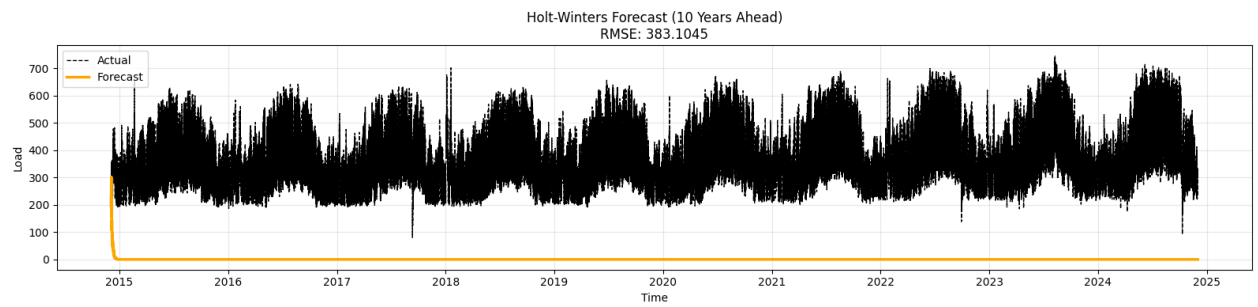


Figure 37: Holt-Winters 10 Year Forecast

1 MONTH RMSE					
Model Type	OLS (Fitted)	Research Model	Random-Forest	SARIMAX	Holt-Winters
Static Pricing	58.69	58.16	61.15	50.21	55.78
Dynamic Pricing	59.58	67.42	61.18	49.98	—
1 YEAR RMSE					
Model Type	OLS (Fitted)	Research Model	Random-Forest	SARIMAX	Holt-Winters
Static Pricing	75.62	58.30	19.08*	Too slow to compute	103.58
Dynamic Pricing	66.75	67.64	18.99*	Too slow to compute	—
10 YEAR RMSE					
Model Type	OLS (Fitted)	Research Model	Random-Forest	SARIMAX	Holt-Winters
Static Pricing	70.10	58.52	29.07*	Too slow to compute	383.10
Dynamic Pricing	61.39	68.56	29.07*	Too slow to compute	—

Figure 38: Condensed Model Results

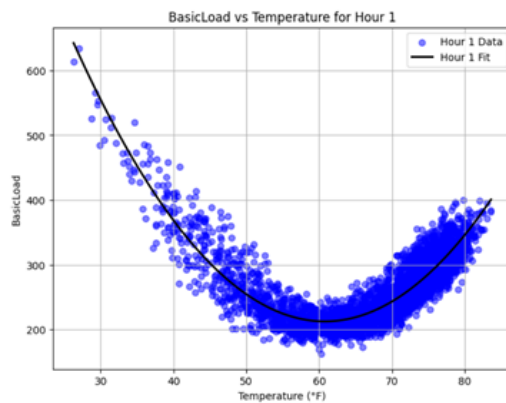


Figure 39: Temperature Variable Research Graph

Appendix 4: Patent Research, Safety and Risk Tables

Patent Research

There were no patents that directly related to the project's scope. Every technique, model, and tool utilized was derived from open-source libraries and publicly accessible scholarly research. No proprietary technology was created or violated.

Safety Considerations

Real-time control systems and physical hardware were not used in this project. As a result, neither participants nor stakeholders faced any physical risks or safety hazards.

Risk Assessment

Basic validation was used to address the main risks, which were associated with data quality and model reliability.

Appendix 5: Individual Contribution Statements

Josiah Jackson:

https://github.com/ijackson3465/Lakeland_Electric

- Communication lead – responsible for scheduling meetings and maintaining contact with everyone including the sponsors. Helped with each presentation and report throughout the entire course.
- Technical lead – responsible for code and model creation.
- EDA – performed a lot of EDA to initially understand weather data and then understanding/utilizing both the data set provided by Lakeland Electric and the research information.
- Models created:
 - o Sensitivity analysis – built off Nick's research and coded the sensitivity analysis using the research model. Created sensitivity analysis visuals for HDD, CDD, windspeed, Price (static and dynamic), population, and GDP (income) to visualize how different values for the elasticities affect load. I also was responsible for updating the research model to adjust correctly for seasonality via daily/hourly adjustments.
 - o OLS model fit to Lakeland Data – Created the model for determining the elasticities by fitting the model directly to our Lakeland data instead of using research elasticities.
 - o Rolling windows forecasting evaluation – performed rolling windows for 1 month, 1-year, 10-year forecast timeframes to identify the optimal window size to forecast.

- Created SARIMAX model to utilize more advanced time series model aimed at accounting for seasonality and exogenous variables. Successfully obtained 1 month forecast, but despite best attempts couldn't get the 1-year or 10-year forecast to work due to limited computational power.
- Created Holt-Winters forecasting model to use as a baseline for seeing how only using past load to predict future load would work.
- Model Evaluations:
 - I created all the visuals shown in Appendix 3 along with interpreting/comparing the models mentioned above.

Nicholas Ricci:

I worked together during regular meetings with Wenslo and Jackson to develop the initial forecasting model (Figure 2 of Appendix 2) as well as working on researching the papers behind the model.

The finalized proof-of-concept forecasting equation expands and builds on the initial forecasting model. The finalized proof-of-concept forecasting equation formed the basis of all our analysis and was contained in the foundational Excel file I created for our project, which compiled all pertinent academic and industry research gathered over the semester into a single, well-organized hub.

In order to try to find a common ground in variables and the elasticity ranges for important variables like price, GDP, population, HDD, CDD, and wind speed, I gathered research from a large number of research papers about the equations used in the papers and developed the full logarithmic forecasting model.

I studied major climate models like GFS, NAM, NBM, and HRRR to inform our methodology (Appendix 6E) and thoroughly researched Itron, Lakeland Electric's current model (Appendix 6D). In order to support the model's structure and elasticity ranges, I also compiled and sourced scholarly articles from Google Scholar and numerous other reliable sources. Additionally, I compiled Lakeland Electric's company research, which included a number of links pertaining to their social media accounts and financial statements (Appendix 6F). Additionally, you can find the citations for research papers 1 through 25 that I used to create the equations (Appendix 6A).

Based on feedback, I updated and modified the problem statement and customer needs to better represent Lakeland Electric's objectives. In addition, I included the previously absent details about the project's limitations, background, and industry norms.

I helped keep the team organized, communicated clearly and promptly, and was always early for meetings.

I frequently went to see the professor during office hours to get important information that I shared with the team to improve our work. The collection of those crucial insights was essential to the project's success because without them, many essential components would have been missing.

Comprehensive Excel File: [IDC4943 Capstone 2 Research Hub For Models.xlsx](#)

Samuel Hesse:

I have worked on the technical research, group coordination, and documentation efforts of our project on building load and revenue models tailored for Lakeland Electric.

Data Preparation Strategy: At the start of the semester, I worked on defining how to split and organize the dataset so it could be effectively used in the base temperature part of the project. This was important for organizing the data into hourly splits with all the different types of data in one place to work from.

Temperature Variable Research: I was responsible for researching the role of temperature, specifically heating and cooling degree days in electricity load forecasting models. I investigated why 65°F is commonly used as the baseline and looked into if this applies to Lakeland. Based on our Lakeland specific data, I found that the baseline temperature fluctuates between 61°F and 67°F depending on the time of day, which may impact the accuracy of standard models when applied locally. The example can be found in Appendix 3 (Appendix 3 Figure 39).

Other Weather Features: I also worked on trying to compare load to the other weather features such as humidity, temperature heat index, windchill, and rainfall. The reason for starting with these correlations was to try to find correlations between these weather features and load for our math formula. As a group we did not end up following through with these correlations I found. Instead, we went with the research route using elasticities from research papers and other models to formulate our math model.

Problem Statement and Customer Needs: I worked with my team to draft, revise, and finalize our project's problem statement and customer needs. This involved several iterations to ensure that our documentation aligned with both the scope of the project and the expectations of our sponsor.

Team Coordination and Task Delegation: I played a leadership role in organizing team meetings, managing workflow, and delegating tasks to make sure team members knew what they were supposed to do. I also helped prepare visual and written components for our class presentations and the panel presentation.

Report Contributions: I contributed to the formal project report by writing sections related to the problem statement and customer objectives. This included wording our customer

objectives and problem statement in a way that reflects the project and what our sponsor wants.

Wenslo Dort:

I collaborated with Nick and Jackson at the start of the project to help create the first load forecasting equation, which served as the foundation for our subsequent modeling attempts. Figure 2 in Appendix 2 provides more information on this equation. Finding and reviewing pertinent scholarly and commercial publications on energy forecasting was one of my main duties. These resources were crucial in helping us develop our knowledge of the main factors affecting the demand for electricity and in making sure that our work was based on tried-and-true techniques. I looked for research that examined how factors like GDP, population growth, weather patterns, and electricity prices affected residential electricity use. Our choices regarding the model's structure and input prioritization were influenced by these papers. Beyond research, I participated in early brainstorming sessions where we evaluated different model configurations and discussed the practical feasibility of applying certain assumptions from the literature to our specific use case. I contributed to conversations around the elasticity of demand in response to pricing and weather variables, helping to identify reasonable ranges based on the studies we reviewed. These efforts ensured our initial model reflected both theoretical soundness and contextual relevance for Lakeland Electric's customer base. My contribution provided a research-driven foundation that supported the analytical direction of the project and helped align our model development with real-world forecasting practices. I consistently showed up and took part in the team's collaborative work sessions for the duration of the project. As the project progressed, I met with Nick and Jackson on a regular basis to review our progress, share ideas, and coordinate our approaches. I participated in the discussions that determined our course, whether we were improving the forecasting formula, going over recently acquired data, or assessing research findings. My regular attendance made sure I was informed of all significant decisions and enabled me to make a significant contribution to conversations regarding research integration and model improvement. These sessions were crucial for assigning tasks as well as for jointly evaluating the practical and technical ramifications of our modeling assumptions.

Maria Valencia:

At the outset of the project, I participated in the initial exploration of the dataset and was involved in the early communications with our sponsor to clarify expectations and understanding of our project's scope. These early efforts helped our team establish a foundational understanding of the data, but this part was mainly a group effort as a whole. Then I also tried to assist with early concepts of our linear models, but then this aspect was then passed to other team members to focus on. After Josiah conducted an in-depth

exploratory data analysis (EDA), I extended this work by contributing additional visualizations (As can be seen in Appendix 6I pages 44 to 94). I also contributed to Appendix 6J which shows a python notebook involving the use of Random Forest buckets. I created heatmaps to assess correlations between our initial variables and developed various relationship graphs to explore interactions among key features. I also performed outlier detection and initiated early feature engineering to enhance model inputs. I built the first Random Forest model using our preliminary set of features, which achieved an accuracy of 88%. I also generated the initial feature importance plot to guide future variable selection. To improve model performance, I conducted a grid search and produced elbow plots to determine the optimal number of estimators for the Random Forest algorithm. Following a change in our dataset to a final cleaned version, I developed a new set of Random Forest models. These models predicted outcomes over various time horizons—monthly, six-month, one-year, five-year, and ten-year periods—across both dynamic and non-dynamic pricing models. Part of these models looked out into the future while some of them limited the dataset so we could evaluate the performance of the model on past actuals vs our predictions. I further created additional model versions using logged transformations of the variables for each model and timeframe for better performance and interpretability. To align with the evaluation methods used by my teammates, I shifted from using accuracy as a metric to Root Mean Squared Error (RMSE), which allowed for more consistent comparisons across models used in this project, made by other team members. While most of the heavy preprocessing work had already been handled by Josiah, I ensured any remaining light data preparation steps were properly completed before modeling for the models I was in charge of developing. For our capstone showcase, I took the lead on designing and producing the project brochures. Although I don't typically engage in artistic tasks, I dedicated significant time to creating a visually appealing and informative brochure that clearly communicated our project's objectives and findings. I also managed the physical printing and hand-folding of each brochure for presentation day. However for what ended up being included in the final state of our project, it really only encompassed that I oversaw the creation and application of the Random Forest model, which was the project's machine learning component. I was in charge of feature selection, data input preprocessing, and Random Forest model parameter tuning, and made the brochure for the capstone showcase.

Jackson Marulis:

I collaborated closely with Nick and Wenslo at the beginning of the project to help lay the groundwork for our initial residential load forecasting model which can be seen in Appendix 2 Figure 2. Finding and evaluating pertinent scholarly and commercial research publications that could guide our modeling approach was one of my main duties. Understanding the effects of important variables on electricity demand, including population trends, weather variability, electricity pricing, and economic indicators like GDP,

was the main goal of this study. The knowledge gathered from this literature made sure that our modeling approach was based on tried-and-true techniques and shaped our comprehension of the dynamics affecting residential energy use. The papers I came across helped us make decisions regarding the theoretical framework of our forecasting efforts, model structure, and variable selection. I actively participated in the team's early brainstorming sessions in addition to doing literature research, where we talked about the implications of different modeling strategies and how well some scholarly approaches could be applied in real-world scenarios for Lakeland Electric. In order to assess which configurations would most accurately represent the clientele and usage trends, I also took part in discussions regarding demand elasticity and the viability of various parameter assumptions. I stayed actively involved in the project and met with Nick and Wenslo on a regular basis to discuss updates, monitor progress, and make sure our efforts were coordinated. These sessions made it possible for us to stay on course and made sure my suggestions were incorporated into the creation of our modeling framework.

Jovan Ortiz:

By helping Maria review and assess the Random Forest model over its many iterations, I contributed to its development. I reviewed intermediate outputs, confirmed results, and participated in discussions that examined the effects of particular variables on predictive accuracy, even though I wasn't directly involved in programming or training the model. During this process, I provided helpful criticism on how the model could more accurately represent the main goals of our forecasting task, like striking a balance between interpretability and performance. I assisted Maria in evaluating whether the patterns the model found were consistent with more general trends seen in the data, as well as how the inclusion or exclusion of particular features might affect the model's output. We were able to identify any possible discrepancies and make sure the model remained conceptually sound thanks to this feedback loop. Along with these initiatives, I participated in a number of planning sessions where we talked about the project's course and potential future research to improve our findings. I helped establish expectations for the types of follow-up work that might be required, like more data preprocessing, model validation, or investigating different machine learning techniques. I also took part in conversations regarding the overall schedule, assisting in determining whether our deliverables and milestones were on schedule or required modification in light of our progress. While my role was largely supportive, my contributions added value by reinforcing quality control, helping steer the project toward its goals, and ensuring team alignment during key phases of development and planning.

Appendix 6: Additional Documents, Important Sources, and Python Code

Appendix 6A: Citations

1) Tariff Document:

Lakeland Electric. (2024). *Residential Service Rate Schedules RS, RSX-1, and RSD*. Lakeland Electric Tariff Documents. Retrieved from the Lakeland Electric utility services on December 1st, 2024.

2) ITRON Information:

Itron. *Itron Energy Forecasting and Analytics*. Retrieved November 16, 2024, from <https://na.itron.com/>

3) Research Paper 1:

Kavaklioglu, K. (2011). Modeling and prediction of Turkey's electricity consumption using Support Vector Regression. *Energy*, 36(5), 2671–2675.
<https://doi.org/10.1016/j.energy.2011.02.051>

4) Research Paper 2:

Arora, S., & Taylor, J. W. (2020). Forecasting electricity smart meter data using conditional kernel density estimation. *2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, 1–6.
<https://doi.org/10.1109/SmartGridComm47815.2020.9302903>

5) Research Paper 3:

Li, M., Cai, W., Gao, X., & Jin, C. (2024). Exploring the determinants of residential electricity consumption in China using a machine learning approach. *Heliyon*, 10(6), e28288.
<https://doi.org/10.1016/j.heliyon.2024.e28288>

6) Research Paper 4:

Csereklyei, Z., & Stern, D. I. (2012). Global energy use: Decoupling or convergence? *Energy Policy*, 51, 633–641. <https://doi.org/10.1016/j.enpol.2012.09.024>

7) Research Paper 5:

Xu, W., Li, J., & Zheng, C. (2024). Price elasticity of residential electricity demand: Evidence from Chinese provincial panel data. *Sustainability*, 16(8), 3321.
<https://doi.org/10.3390/su16083321>

8) Research Paper 6:

Krieger-Boden, C., & Schumacher, I. (2022). Energy prices and household consumption behavior: Evidence from German micro data. *Econstor Discussion Papers*.
<https://www.econstor.eu/handle/10419/249570>

9) Research Paper 7:

Ahmad, T., Chen, H., & Guo, Y. (2018). A review on demand side management: Electrical power system optimization models. *Journal of Cleaner Production*, 205, 987–1000.
<https://doi.org/10.1016/j.jclepro.2018.09.137>

10) Research Paper 8:

Simmhan, Y., Kumbhare, A., Cao, B., & Prasanna, V. K. (2013). An analysis of heterogeneity and information hiding in demand response management. *2013 IEEE International Conference on Smart Grid Communications (SmartGridComm)*, 770–775.
<https://doi.org/10.1109/SmartGridComm.2013.6688034>

11) Research Paper 9:

Csereklyei, Z. (2017). The Impact of Economic Structure on Energy Use: Evidence from Australia. *Australian National University – Arndt-Corden Department of Economics*.
https://ajrc.crawford.anu.edu.au/sites/default/files/events/attachments/2017-09/csereklyei_abstract.pdf

12) Research Paper 10:

Auffhammer, M., & Rubin, E. (2018). Climate Change, Mortality, and Adaptation: Evidence from Annual Fluctuations in Weather in the U.S. *National Bureau of Economic Research Working Paper No. 23483*.
https://www.nber.org/system/files/working_papers/w23483/revisions/w23483.rev2.pdf

13) Research Paper 11:

U.S. Energy Information Administration (EIA). (2014). *Price Elasticities for Energy Use in*

Buildings of the United States.

https://www.eia.gov/analysis/studies/buildings/energyuse/pdf/price_elasticities.pdf

14) Research Paper 12:

Caldwell, J. H. (2020). *Price Elasticity and Electricity Rate Design*. Pacific Energy Institute White Paper. <https://www.pacificenergyinstitute.org/wp-content/uploads/2020/02/Caldwell-Price-Electricity-and-Electricity-Rate-Design.pdf>

15) Research Paper 13:

Zachmann, G. (2016). Electricity wholesale market prices and demand response. *Robert Schuman Centre for Advanced Studies Research Paper No. RSCAS 2016/25*. https://cadmus.eui.eu/bitstream/handle/1814/40870/RSCAS_2016_25.pdf

16) Research Paper 14:

Weron, R. (2010). Modeling and forecasting electricity loads and prices: A statistical approach. *arXiv preprint arXiv:0912.1015*. <https://arxiv.org/pdf/0912.1015>

17) Research Paper 15:

Amoako, J., & Osei, V. (2011). Electricity demand estimation in Ghana. *Energy and Power Engineering*, 3(5), 537–543. <https://www.scirp.org/journal/paperinformation?paperid=4960>

18) Research Paper 16:

Hinman, K., & Hickey, B. (2009). Load Forecasting: The Key to Demand Side Management. *Illinois State University Undergraduate Research*. <https://irps.illinoisstate.edu/downloads/research/student/LoadForecastingHinman-HickeyFall2009.pdf>

19) Research Paper 17:

Hyndman, R. J., & Fan, S. (2010). Density forecasting for long-term peak electricity demand. *IEEE Transactions on Power Systems*, 25(2), 1142–1153. <https://www.robjhyndman.com/papers/Elasticity2010R1.pdf>

20) Research Paper 18:

Islam, S., et al. (2022). Climate change impacts on electricity demand: A review. *Climate*, 10(10), 142. <https://doi.org/10.3390/cli10100142>

21) Research Paper 19:

Jacob, K. (n.d.). Residential demand for electricity in Florida: A microeconomic analysis. *Public Utility Research Center, University of Florida*.
https://bear.warrington.ufl.edu/centers/purc/docs/papers/7604_Jacob_Demand_for_Electricity.pdf

22) Research Paper 20:

Florida Reliability Coordinating Council (FRCC). (2023). *Ten-Year Site Plan Resource and Load Report*. https://www.floridapsc.com/pscfiles/website-files/PDF/Utilities/Electricgas/TenYearSitePlans/2023/FRCC_RLRP.pdf

23) Research Paper 21:

Holt, J. R., & Razghandi, E. (2020). Machine Learning-Based Load Forecasting for Smart Grid Applications. *2020 IEEE Global Communications Conference (GLOBECOM)*, 1–6.
<https://www.eecs.ucf.edu/~turgut/Research/Publications/Download/Razghandi-2020-GLOBECOM.pdf>

24) Research Paper 22:

Florida Municipal Power Agency (FMPA). (2020). *All-Requirements Project Load Forecast*. <https://portal.fmpa.com/wp-content/uploads/2021/03/ARP-Load-Forecast-2020.pdf>

25) Research Paper 23:

Liu, Y., et al. (2022). Residential electricity consumption forecasting based on time series decomposition and CNN-LSTM model. *2022 International Conference on Power System Technology (POWERCON)*. <https://ieeexplore.ieee.org/document/9888131>

26) Research Paper 24:

Auffhammer, M., Baylis, P., & Hausman, C. H. (2017). Climate change is projected to have severe impacts on the frequency and intensity of peak electricity demand across the United States. *Climatic Change*, 148(2), 355–368. <https://doi.org/10.1007/s10584-011-0297-0>

27) Research Paper 25:

Gupta, E. (2014). *Understanding the factors determining residential electricity consumption in India*. Working Paper No. 14-08, Indian Statistical Institute.
<http://www.isid.ac.in/~epu/wp-content/uploads/2014/09/dp14-08.pdf>

28) Research Paper 26:

Emenekwe, C. C., & Emodi, N. V. (2022). *Analysis of electricity demand elasticities in Nigeria using ARDL and GMM techniques*. *Climate*, 10(10), 142.
<https://www.mdpi.com/2225-1154/10/10/142>

29) Research Paper 27:

Atalla, T., & Hunt, L. C. (2016). *Analysis of the long-run relationship between electricity consumption and economic growth in GCC countries*. *Energy Economics*, 59, 374–382.
<https://doi.org/10.1016/j.eneco.2016.07.027>

30) Research Paper 28:

Boogen, N., Datta, S., & Filippini, M. (2013). *Regional long-term electricity demand and supply scenarios: A Swiss case study*. *Energy Economics*, 36, 637–646.
https://folia.unifr.ch/documents/318587/files/filippini_EE_2013.pdf

31) Research Paper 29:

Almuhtady, A., Roeser, F., Rashed, E., & Meyer, R. (2019). *Short-term electric load forecasting with multiple weather variables using regression and neural network models*. *Energy, Sustainability and Society*, 9(1), 24.
<https://link.springer.com/content/pdf/10.1186/s13705-019-0224-1.pdf>

32) Research Paper 30:

Fan, Y., Xiao, L., Liu, J., et al. (2019). *Short-term electric load forecasting based on multivariable correlation analysis and hybrid model in China*. *Energy*, 175, 1106–1119.
<https://www.sciencedirect.com/science/article/pii/S0360544218324101>

33) Research Paper 31:

Idsø, J., Huseby, R. B., & Sægrov, S. (2024). *The impact of weather conditions on household electricity consumption in Norway*. *Sustainability*, 16(8), 3321.
<https://www.mdpi.com/2071-1050/16/8/3321>

Appendix 6B: Lakeland Electric Relevant 10 Year Plan Information

“3.0 Forecast of Electric Demand and Energy

Annually, Lakeland Electric (LE) develops a detailed short-term (1 year) electric load and energy forecast for budget purposes and short-term operational studies. An annual long-term (10 years) forecast is developed for the Utility’s long-term planning studies including LE’s Ten Year Site Plan (TYSP).

Sales and customer forecasts of monthly data are prepared by rate classification. Separate forecast models are developed for inside and outside the City of Lakeland corporate limits for the Residential, Commercial, Industrial and Other (municipal departments and outdoor lighting) rate classifications. Monthly forecasts are summarized annually using fiscal period ending September 30th

for the short-term budget forecast and by calendar year for long-term studies and reporting.

Lakeland Electric uses MetrixND, an advanced statistical forecasting software tool, developed by Itron, to assist with the development of LE’s number of customers, energy and demand forecasts. Lakeland Electric uses MetrixLT, another Itron software tool, which integrates with MetrixND to develop the long-term system hourly load forecast.

The modeling techniques used to generate the forecasts include multiple regression, study of historical relationships and growth rates, trend analysis, and exponential smoothing. Lakeland Electric utilizes Itron’s Statistically Adjusted End-Use (SAE) econometric modeling approach for the residential and commercial sectors. The SAE approach is designed to capture the impact of changing end-use saturation and efficiency trends, by building type, as well as economic conditions on long-term residential and commercial energy sales and demand.

Many variables are evaluated for the development of the forecasts. The variables that have proven to be significant and are included in the forecasts are weather, gross regional product, disposable personal income per household, persons per household, number of households, local population, electricity price, building type, appliance saturation and efficiency. Binary variables are used to explain outliers in historical billing discrepancies, trend shifts, monthly seasonality, rate migration between classes and other issues that could affect the accuracy of forecast models.

Weather variables

Heating and cooling degree days are weather variables that attempt to explain a customer’s usage behavior as influenced by either hot or cold weather. Heating Degree Days (HDD) occur when the average daily temperature is less than Lakeland Electric’s established base temperature of 65 degrees Fahrenheit. Cooling Degree Days (CDD) occur when the average daily temperature is greater than 65 degrees. The formulas used to determine the number of degree days are:

$$HDD = \text{Base Temperature (65)} - \text{Average Daily Temperature}$$

CDD = Average Daily Temperature – Base Temperature (65)

These HDD and CDD variables are used in the forecasting process to correlate electric consumption with weather. The HDD and CDD variables are weighted to capture the impacts of weather on revenue from monthly billed consumption.

Lakeland Electric uses weather data from its own weather stations, which are strategically placed throughout the electric service territory to provide the best estimate of overall temperature for the Lakeland Electric service area.

The most recent 20 years of historical normal weather is used as an input into the sales forecast models.

Normal peak-producing weather is also developed using historical 20-years weather. A weighted average of temperatures on both the day of historical monthly peak and day prior to peak is used to create the HDD and CDD variables.

Economic and demographic variables

The economic and demographic projections used in the forecasts are purchased from Moody's Analytics.

Price variables

A real price forecast by month and rate class is created based on Lakeland Electric historical price data, projections from the Lakeland Electric Rates and Fuel teams, the U.S. Energy Information Administration (EIA) Annual Energy Outlook (AEO) forecasted price of electricity, historical and projected Net Energy for Load, and the projected Consumer Price Index. The 12month moving average of projected real price of electricity is the price variable used in the sales and demand SAE models. *Structural Indices*

The end-use saturation and efficiency indices used in the models are purchased from Itron. Itron's Energy Forecasting Group (EFG) offers end-use data services and forecasting support.

EFG's projections are based on data derived from the EIA's AEO forecast for the South Atlantic

Census Division. Itron is also contracted to further calibrate the indices based on Lakeland Electric's service area using average square feet by building type for the Commercial Sector and average use by dwelling type for the Residential Sector.

Lakeland Electric reviews the forecasts for reasonableness, compares projections to historical patterns, and modifies the results as needed using informed judgment.

Historical monthly data is available and is analyzed for the 20-year period. Careful evaluation of the data and model statistics is performed; this often results in most models being developed using less than a 10-year estimation period.

Lakeland Electric currently does not have any specific energy savings goals through Demand Side Management (DSM) programs; therefore, Lakeland Electric does not assume any deductions in peak load for the forecast period.

Lakeland Electric forecasts the number of monthly electric accounts for the following categories and subcategories:

- Residential, Inside and Outside City Limits
- Commercial, Inside and Outside City Limits
- Industrial, Inside and Outside City Limits
- Other, Inside and Outside City Limits

Lakeland Electric's Energy Sales Forecast is the sum of the following forecasts:

- Residential, Inside and Outside City Limits
- Commercial, Inside and Outside City Limits
- Industrial, Inside and Outside City Limits
- Other, Inside and Outside City Limits

Residential Energy Sales Forecast

The Residential energy sales forecast is developed using the Statistically Adjusted EndUse (SAE) econometric modeling approach.

The residential sales models are estimated with historical monthly energy sales data. They are average use models based on the following equation:

$$AvgUse_{y,m} = b_0 + b_1 XCool_{y,m} + b_2 XHeat_{y,m} + b_3 XOther_{y,m} + \epsilon_{y,m}$$

Where $XCool_{y,m}$, $XHeat_{y,m}$ and $XOther_{y,m}$ are explanatory variables constructed from weather data, end use equipment efficiency and saturation trends, economic and demographic data, dwelling type (single family, multi family or mobile home) and square footage." * See attached 10 year Plan (Lakeland Electric) for additional information

Appendix 6C: Lakeland Electric Relevant Tariff Document Information

"RATE SCHEDULE - RS RESIDENTIAL SERVICE

Available:

In all territory served by Lakeland Electric.

Applicable:

To all electric service provided to single family homes, mobile homes, apartments, condominiums, or cooperative apartment buildings where such energy usage is exclusively for residential purposes subject to the following requirements:

1. 100% of the energy used is exclusively for the customer's benefit.

2. None of the energy is used in any endeavor which sells or rents a commodity or provides service for a fee.

3. Each point of delivery will be separately metered and billed.

4. A responsible legal entity is established as the customer to whom a bill can be rendered.

5. Beginning January 1, 2016, new solar electric systems interconnected with

Lakeland Electric shall take service under Rate Schedule Residential Service Demand (RSD). Existing customers as of this date may maintain service under this rate scheduled through December 31, 2025.

Character of Service:

A-C; 60 Hertz; single-phase, 3-wire; 120/240 volts or 120/208 volts.

Limitation of Service:

Standby service or resale not permitted under this rate schedule.

Net Rate per Month:

Customer Charge: \$14.00

Energy Charge: 0 to 1,000 kWh \$0.05392 per kWh

1,001 to 1,500 kWh \$0.06161 per kWh

Above 1,500 kWh \$0.06816 per kWh

Minimum Bill:

Customer charge, plus Adjustments.

Adjustments:

Fuel charge, as contained in Schedule BA-1

City Utility Tax or Surcharge, taxes, surcharges, and fees as contained in Schedule BA-2

Environmental Compliance Cost Charge as contained in Schedule BA-3

Smart Grid Project Implementation as contained in Schedule BA-5

Payment:

Net bills are due when rendered and are delinquent thirty (30) days after the billing date.

Terms and Conditions:

1. All service hereunder will be supplied at one location through one point of delivery and measured through one meter.
2. Service hereunder is subject to the rules and regulations for electric service as adopted by Lakeland Electric from time to time and on file with the City Clerk.

RATE SCHEDULE RSX-1 RESIDENTIAL SERVICE SHIFT TO SAVE

OPTIONAL TIME-OF-DAY

Available:

In all territory served by Lakeland Electric.

Applicable:

To all electric services provided to single family homes, mobile homes, apartments, condominiums, or cooperative apartment buildings where such energy usage is exclusively for residential purposes subject to the following requirements.

1. 100% of the energy used is exclusively for the customers' benefit.
2. None of the energy is used in any endeavor which sells or rents a commodity or provides a service for a fee.
3. Each point of delivery will be separately metered and billed.
4. A responsible legal entity is established as the customer to whom a bill can be rendered.
5. After January 1, 2016, service is no longer available for a customer with a solar electric system interconnected with Lakeland Electric. Service shall be moved to Rate Schedule Residential Service Demand (RSD).

Character of Service:

A-C; 60 Hertz; single-phase 3-wire; 120/240 volts or 120/208 volts.

Limitation of Service:

Resale not permitted under this rate schedule.

Net Rate per Month:

Customer Charge: \$ 14.00
Energy Charge:
On-Peak: \$0.13709 per kWh

Off-Peak: \$0.01544 per kWh

Definitions of the Time-of-Day Periods:

All time periods are stated in prevailing time.

Summer Winter

On-Peak Hours Apr. 1 - Oct. 31 Nov. 1 - March 31

(Monday-Friday) 12:01 PM - 9:00 PM 6:01 AM 10:00 AM, and, 6:01 PM - 10:00 PM

Off-Peak Hours: All other weekday hours, and all hours on Saturdays, Sundays, New Year's Day, Memorial Day, Independence Day, Labor Day, Thanksgiving Day, and Christmas Day shall be off-peak.

Minimum Bill:

Customer Charge, plus Adjustments.

Adjustments:

Fuel charge, as contained in Schedule BA-1

City Utility Tax or Surcharge, taxes, surcharges, and fees as contained in Schedule BA-2

Environmental Compliance Cost Charge as contained in Schedule BA-3

Smart Grid Project Implementation as contained in Schedule BA-5

Payment:

Net bills are due when rendered and are delinquent thirty (30) days after the billing date.

Terms and Conditions:

1. All service hereunder will be supplied at one location through one point of delivery and measured through one meter.
2. Service hereunder is subject to the rules and regulations for electric service as adopted by Lakeland Electric from time to time and on file with the City Clerk.

3. Customers who select to take service hereunder and subsequently terminate service shall be prohibited from returning to service hereunder for twelve (12) months unless approved by Lakeland Electric.

RATE SCHEDULE RSD RESIDENTIAL SERVICE DEMAND

RESIDENTIAL PEAK DEMAND

Available: In all territory served by Lakeland Electric

Applicable:

To all electric services provided to single family homes, mobile homes, apartments, condominiums, or cooperative apartment buildings where such energy usage is exclusively for residential purposes subject to the following requirements:

1) Residential customer with solar electric systems interconnected with Lakeland Electric shall take service under this rate schedule either:

a) Beginning January 1, 2016, for new customers and existing customers under Rate Schedule Residential Service Shift to Save Optional Time-of-Day (RSX-1), or,

b) January 1, 2026, for customers who were on Rate Schedule Residential Service (RS) before

January 1, 2016.

2) Residential customer with solar electric systems interconnected with Lakeland Electric shall receive a Value of Solar Credit beginning October 1, 2018.

Otherwise, service hereunder is available at the customer's option.

3) One-hundred percent (100%) of the energy used is exclusively for the customer's benefit.

4) None of the energy is used in any endeavor which sells or rents a commodity or provides service for a fee.

5) Each point of delivery will be separately metered and billed.

6) A responsible legal entity is established as the customer to whom a bill can be rendered.

Character of Service: A-C; 60 Hertz; single-phase 3-wire, 120/240 volts or 120/208 volts.

Limitation of Service: Standby service or resale not permitted under this rate schedule.

Net Rate Per Month:

Customer Charge: \$14.00

Demand Charge: \$6.49 per kW of Billing Demand

Value of Solar Credit: \$0.92 per kW of Billing Demand

Energy Charge: \$0.02592 per kWh

Definition of the Time-of-Day On-Peak Period:

All time periods are stated in prevailing time.

Summer Winter

On-Peak Hours Apr. 1 - Oct. 31 Nov. 1 - March 31

(Monday-Friday) 1:01 PM - 8:00 PM 6:01 AM - 10:00 AM

Off-Peak Hours All other weekday hours, and all hours on Saturdays, Sundays, New Year's Day, Memorial Day, Independence Day, Labor Day, Thanksgiving Day, and Christmas Day shall be off-peak.

Minimum Bill:

Customer charge, plus Adjustments.

Adjustments:

Fuel charge, as contained in Schedule BA-1

City Utility Tax or Surcharge, taxes, surcharges, and fees as contained in Schedule BA-2

Environmental Compliance Cost Charge as contained in Schedule BA-3

Smart Grid Project Implementation as contained in Schedule BA-5

Payment:

Net bills are due when rendered and are delinquent thirty (30) days after the billing date.

Determination of Billing Demand:

The billing demand for the month shall be the maximum 30-minute integrated kilowatt on-peak demand in the month.

Where charges specified in a rate schedule are based upon the measured maximum 30minute integrated demand, it is intended that such demand shall fairly represent the capacity which Lakeland Electric is required to stand ready to supply. In case of installations which use this service in a manner such that measurement over a 30-minute interval does not result in a fair or equitable measure of the supply capacity required to service the customer's load, then the measured demand may be adjusted taking into account the known character of use and the rating data of the equipment connected, or from special tests, the intent being that the demand so determined shall fairly represent the customer's capacity requirement.

In cases where Lakeland Electric elects to use connected load instead of demand measurement, as the method for determining demand, it will take into account the known character of use and the rating data of the equipment connected, the intent being that the demand so determined shall fairly represent the customer's capacity requirement.

Terms and Conditions:

1. All service hereunder will be supplied at one location through one point of delivery and measured through one meter.
2. Service hereunder is subject to the rules and regulations for electric service as adopted by Lakeland Electric from time to time and on file with the City Clerk."

* See Attached Tariff Document for further information.

Appendix 6D: ITRON Research Links

What Itron is about:

Itron is a proven global leader in energy, water, smart city, IIoT and intelligent infrastructure services. For utilities, cities and society, we build innovative systems, create new efficiencies, connect communities, encourage conservation and increase resourcefulness. By safeguarding our invaluable natural resources today and tomorrow, we improve the quality of life for people around the world.

Company website: <https://na.itron.com/>

Stock market information:

- <https://investors.itron.com/stock-information>
- <https://companiesmarketcap.com/itron/revenue/>
- <https://investors.itron.com/node/23631/pdf>
- <https://investors.itron.com/static-files/af39412c-220f-4c29-94cf-eda9fcc115d6>

- <https://investors.itron.com/static-files/f81951f0-e7d8-4ea2-b207-2e4effc81c5e>

PDF containing a press release and links to their social media: <https://investors.itron.com/node/24536/pdf>

Itron YouTube Channel: [Welcome to the Active Grid](#)

All about Itron's Distributed Energy Resource Optimizer (Has very nice resources and information pertaining Evs): <https://na.itron.com/products/der-optimizer>

Appendix 6E: Weather Information and Python Code for Obtaining Weather Data.

Weather Research Links:

- <https://www.timeanddate.com/weather/usa/lakeland/historic>
- <https://www.wunderground.com/forecast/us/fl/lakeland>
- <https://www.weather.gov/wrh/timeseries?site=KLAL>
- <https://climatecenter.fsu.edu/climate-data-access-tools/downloadable-data>
- <https://fawn.ifas.ufl.edu/data/>
- <https://graphical.weather.gov/sectors/florida.php>
- <https://www.wunderground.com/history/daily/us/fl/lakeland/KLAL>

Links to Understand GFS, NAM, NBM, HRRR

Comprehensive PDF containing details about GFS, NAM, NBM, HRRR

https://mag.ncep.noaa.gov/docs/NCEP_PDD_MAG.pdf

GFS Links

<https://www.emc.ncep.noaa.gov/officenotes/newernotes/on442.pdf><https://www.emc.ncep.noaa.gov/officenotes/newernotes/on442.pdf>
<https://www.ncei.noaa.gov/products/weather-climate-models/global-forecast> <https://registry.opendata.aws/noaa-gfs-pds/>
https://developers.google.com/earth-engine/datasets/catalog/NOAA_GFS0P25
<https://www.rdocumentation.org/packages/rNOMADS/versions/1.2.0/topics/gfs.model.data>
<https://vlab.noaa.gov/web/gfs/documentation>
<https://microsoft.github.io/AIforEarthDataSets/data/noaa-gfs.html>

<https://stackoverflow.com/questions/42298981/documentation-for-noaa-gfs-output-files> <https://registry.opendata.aws/noaa-gfs-bdp-pds/>
<https://www.ncei.noaa.gov/access/metadata/landingpage/bin/iso?id=gov.noaa.ncdc:C00631>

NAM Links

<https://ams.confex.com/ams/pdfpapers/154114.pdf>
<https://www.ncei.noaa.gov/products/weather-climate-models/north-american-mesoscale> <https://registry.opendata.aws/noaa-nam/>
<https://www.weather.gov/rah/namnomogram>
<https://www.ncei.noaa.gov/access/metadata/landingpage/bin/iso?id=gov.noaa.ncdc:C00630> <http://wxmaps.org/pix/nam.fcst> <https://windy.app/blog/what-is-nam-weather-model-and-how-it-works.html> <https://luckgrib.com/models/nam/>
https://emc.ncep.noaa.gov/emc/pages/numerical_forecast_systems/nam.php
<https://catalog.data.gov/dataset/north-american-mesoscale-forecast-system-nam-12-km2>

NBM Links <https://www.weather.gov/news/200318-nbm32>
<https://vlab.noaa.gov/web/mdl/nbm>
<https://theweatherguy.net/blog/basic-info-about-the-nbm-and-icon-models/> <https://registry.opendata.aws/noaa-nbm/>
https://luckgrib.com/models/nbm_oceanic/
https://nomads.ncep.noaa.gov/txt_descriptions/BLEND_txt.html
<https://vlab.noaa.gov/web/mdl/nbm>
<https://aws.amazon.com/marketplace/pp/prodview-dn3lfyvesyh42#links>

HRRR Links

[https://learningweather.psu.edu/node/90#:~:text=The%20RR%20model%20provides%20data,more%20simply,%20the%20HRRR\).](https://learningweather.psu.edu/node/90#:~:text=The%20RR%20model%20provides%20data,more%20simply,%20the%20HRRR).)
<https://rapidrefresh.noaa.gov/hrrr/>
<https://www.facebook.com/NWSBoulder/videos/hrrr-model-forecast-of-radarreflectivity-one-of-our-forecasting-tools-indicates/2448958835132461/>
<https://windy.app/blog/what-is-hrrr-weather-model-and-how-it-works.html>
<https://journals.ametsoc.org/view/journals/wefo/37/8/WAF-D-21-0151.1.xml>
<https://www.weather.gov/news/200210-rapid-model>
https://repository.library.noaa.gov/view/noaa/53029/noaa_53029_DS1.pdf
<https://www.pivotalweather.com/model.php?m=hrrr>
https://www.tropicaltidbits.com/analysis/models/?model=hrrr®ion=seus&pkg=ref_frz_n

Summaries/Explanations to Understand GFS, NAM, NBM, HRRR

GFS (Global Forecast System):

- A global model run by the US Government (NOAA) and its subsidiary agencies

- Funded by American taxpayers, making its forecast output freely available
- Hydrostatic model, meaning it assumes a constant atmospheric pressure with height
- Runs every 6 hours, predicting conditions out to 16 days, with highest accuracy in the 1-4 day range
- Used by many weather websites and apps

NAM (North American Mesoscale Forecast System):

- A limited-area model, focusing on North America and a portion of the northern Pacific, Arctic, and Atlantic Oceans
- Run by NOAA/NCEP, with a 12-km resolution and a nest covering the continental US at 3-km resolution
- Non-hydrostatic model, allowing for more accurate representation of atmospheric phenomena
- Runs every 6 hours, predicting conditions out to 84 hours (12-km domain) and 60 hours (3-km nest)
- Often overly-aggressive with initiating convection over the US, leading to differences in forecasts compared to other models

NBM (North American Mesoscale Blend):

- An ensemble approach, combining multiple models (including GFS, NAM, and others) to produce a single forecast
- Aims to reduce uncertainty by leveraging the strengths of each individual model
- Not a single model, but rather a blended forecast

HRRR (High Resolution Rapid Refresh):

- A regional model, focusing on the contiguous US and a portion of Canada
- Run by NOAA/NCEP, with a 3-km resolution and hourly initialization
- Non-hydrostatic model, allowing for high-resolution representation of atmospheric phenomena
- Runs every hour, providing rapid updates and detailed forecasts for short-term weather events
- Often used in conjunction with other models, such as GFS and NAM, to improve forecast accuracy

Key Differences Between the Models:

- GFS is a global model, while NAM and HRRR are regional models focused on North America.
- NAM and HRRR are non-hydrostatic models, while GFS is hydrostatic.
- HRRR runs hourly, while GFS and NAM run every 6 hours.
- NBM is an ensemble approach, blending multiple models, while the other models are individual forecasting systems.

Ideas About How GFS, NAM, NBM, and HRRR Account for Seasonality, Time trends, Model Validation, Risk Tolerance, Historical Data/Bias

Seasonality:

- GFS and NAM: As global forecast models, they inherently account for seasonal patterns and cycles in their formulations. However, their performance can vary depending on the season and specific weather phenomena.
- NBM: As a blended model, it incorporates multiple models' outputs, including those that account for seasonality. The NBM's bias correction and weighting scheme help to reduce seasonal biases.
- HRRR (High-Resolution Rapid Refresh): As a high-resolution model, it is designed to capture local and regional weather patterns, including seasonal variations.

Time Trends:

GFS and NAM: They are designed to forecast future weather patterns, incorporating current conditions and trends. Their forecasts are updated regularly, reflecting changes in the atmosphere and evolving weather patterns.

- NBM: By blending multiple models, it captures a range of time scales and trends, from short-term to long-term. The model's weighting scheme helps to reduce biases and improve overall trend representation.
- HRRR: Its high-resolution output allows it to capture rapid changes and trends in weather patterns, making it suitable for short-term forecasting.

Model Validation:

- GFS and NAM: The National Weather Service (NWS) and other organizations regularly validate these models against observational data, such as surface weather

stations, radar, and satellite imagery. This process helps identify biases and areas for improvement.

- NBM: The NWS validates the NBM's performance against a range of observational data, including those mentioned above. The model's bias correction and weighting scheme are designed to minimize errors and improve overall accuracy.
- HRRR: Its high-resolution output makes it challenging to validate directly against observational data. Instead, the NWS and other organizations use ensemble forecasting and other techniques to evaluate its performance.

Risk Tolerance:

- GFS and NAM: As global forecast models, they are designed to provide probabilistic forecasts, which inherently account for uncertainty and risk. The models' ensemble forecasting capabilities help quantify the range of possible outcomes.
- NBM: By blending multiple models, it provides a more comprehensive representation of uncertainty and risk. The model's weighting scheme helps to reduce biases and improve overall risk assessment.
- HRRR: Its high-resolution output allows it to capture localized and rapid changes, which can be useful for risk assessment and decision-making.

Historical Data/Bias:

- GFS and NAM: Historical data and biases are incorporated into their formulations through various techniques, such as model parameterizations and initialization procedures.
- NBM: The model's bias correction and weighting scheme are designed to reduce historical biases and improve overall accuracy. The NBM's blended approach helps to minimize the impact of individual models' biases.

HRRR: Its high-resolution output is designed to capture local and regional weather patterns, which can be influenced by historical conditions and biases. The NWS and other organizations continually evaluate and refine the model's performance to reduce biases and improve accuracy.

Appendix 6F: Lakeland Electric Company Research

Company Website: <https://lakelandelectric.com/>

Facebook Page: <https://www.facebook.com/MyLakelandElectric/>

YouTube Page: [Building the Future](#)

LinkedIn Page: <https://www.linkedin.com/company/lakeland-electric>

Twitter Page: <https://x.com/mylkldelectric>

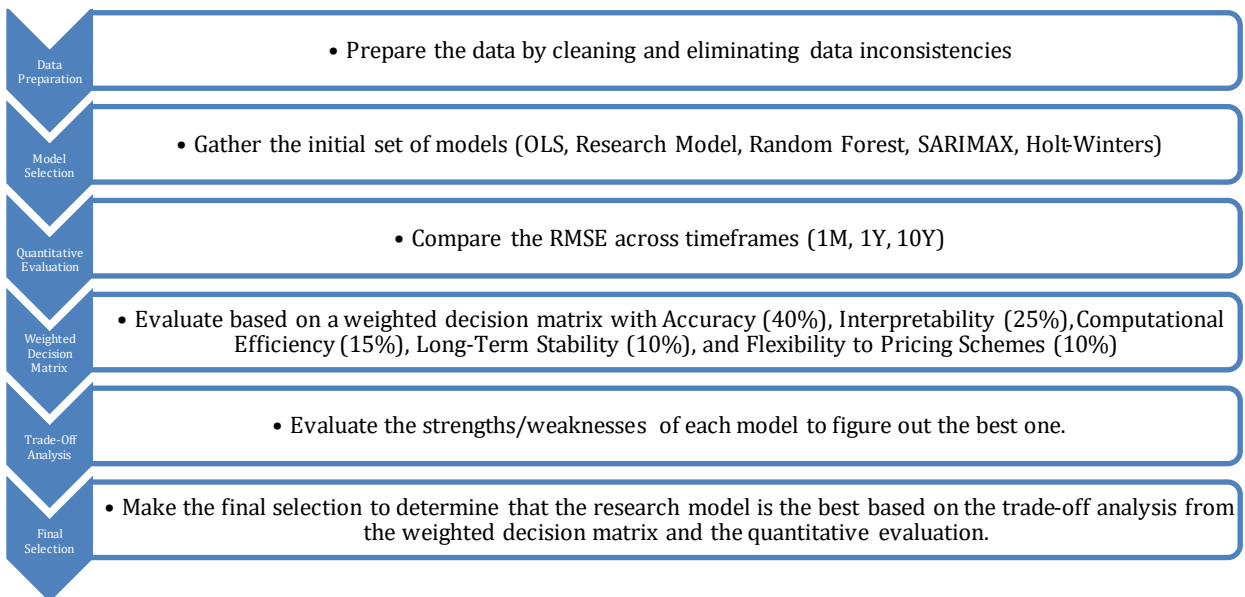
Public Statistics and Rates (Very Useful Website): <https://findenergy.com/providers/lakeland-electric/>

Stock Market Information: <https://finviz.com/quote.ashx?t=LAKE&p=d>

Payment and Pricing Plans (Ease of Access): <https://lakelandelectric.com/billing-and-payment/pricing-plans>

Locational Information: <https://www.zoominfo.com/c/lakeland-electric/58756495>

Appendix 6G: Flowchart of Design Process



Appendix 6H: Weather Python Code/Outputs For Obtaining Data From the Above Sites

```
# Lakeland Electrical Capstone Project

## Potential Data Sources - publicly available

- NOAA or National Weather Association - Free weather data

- U.S Census Bureau or American Community Survey - Census Data
```

- Try to find a Lakeland Regional one
- Energy Information Administration or OpenEI - Energy Industry Standards
- national/state level only

Weather Start

- NWS API - <https://api.weather.gov/>
- Info: <https://www.weather.gov/documentation/services-web-api>
- Lakeland grid -
https://forecast.weather.gov/MapClick.php?x=178&y=130&site=tbw&zmx=&zmy=&map_x=178&map_y=130 = Lat 28.05 N, Lon 81.96 W (Rough Center)
- Lakeland Electrical Coverage -
<https://www.arcgis.com/apps/Viewer/index.html?appid=ffbd05dc2e94406ebcad89e30636877c>

FORECASTS - NEXT 12 HOURS AND NEXT WEEK

```

#!pip install requests import
requests import pandas as pd
from datetime import datetime,
timedelta

# API for Lakeland - chosen rough center of Lakeland Electrical
coverage map lat, lon = 28.05, -81.96 points_url =
f'https://api.weather.gov/points/{lat},{lon}'

# Define headers for the API
request_headers = {
    "User-Agent": "myweatherapp (contact@myweatherapp.com)"
}

# Fetch the grid information response
= requests.get(points_url,
headers=headers) if
response.status_code == 200:
    data = response.json() forecast_url =
data["properties"]["forecast"]
forecast_hourly_url =
data["properties"]["forecastHourly"]

```

```

grid_url =
data["properties"]["forecastGridData"]
print("12-hour Forecast URL:", forecast_url)
# 12-hour forecast    forecast_response =
requests.get(forecast_url, headers=headers)    if
forecast_response.status_code == 200:
forecast_data = forecast_response.json()
forecast_periods =
forecast_data["properties"]["periods"]
# Convert to DataFrame
forecast_df =
pd.DataFrame(forecast_periods)
# Precipitation parsing - only output
the number
forecast_df["probabilityOfPrecipitation"] =
forecast_df["probabilityOfPrecipitation"].apply(    lambda x: x["value"] if x and
x["value"] is not None else 0
)

# Generate a new date column that increments
every two rows    start_date =
datetime.now().date()    forecast_dates = [
(start_date + timedelta(days=i //
2)).strftime("%Y-%m-%d")    for i in
range(len(forecast_df))
]

forecast_df.insert(0, "forecastDate", forecast_dates)

# Columns to keep
forecast_df =
forecast_df[[
"forecastDate", "name", "temperature", "shortForecast",

"probabilityOfPrecipitation", "windSpeed", "windDirection",

"isDaytime", "detailedForecast"

]]

print("\n12-hour Forecast with
Additional Features:")
display(forecast_df)    else:
print("Error fetching 12-hour forecast
data") else:
print("Error fetching information")

```

12-hour Forecast URL: <https://api.weather.gov/gridpoints/TBW/90,104/forecast>

12-hour Forecast with Additional Features:

	forecastDate	name	temperature	shortForecast	probabilityOfPrecipitation	windSpeed	windDirection	isDaytime	detailedForecast
0	2024-11-20	This Afternoon	80	Chance Showers And Thunderstorms	50	9 mph	SW	True	A chance of showers and thunderstorms. Mostly ...
1	2024-11-20	Tonight	55	Slight Chance Rain Showers then Partly Cloudy	20	7 to 12 mph	NW	False	A slight chance of rain showers before 7pm. Pa...
2	2024-11-21	Thursday	67	Sunny	0	12 mph	NNW	True	Sunny. High near 67, with temperatures falling...
3	2024-11-21	Thursday Night	48	Clear	0	8 mph	NW	False	Clear, with a low around 48. Northwest wind ar...
4	2024-11-22	Friday	68	Sunny	0	6 to 13 mph	NW	True	Sunny, with a high near 68. Northwest wind 6 t...
5	2024-11-22	Friday Night	44	Clear	0	5 to 8 mph	NNW	False	Clear, with a low around 44. North northwest w...
6	2024-11-23	Saturday	66	Sunny	0	7 mph	NNW	True	Sunny, with a high near 66. North northwest wi...
7	2024-11-23	Saturday Night	46	Clear	0	1 to 5 mph	N	False	Clear, with a low around 46. North wind 1 to 5...
8	2024-11-24	Sunday	72	Sunny	0	2 mph	ENE	True	Sunny, with a high near 72. East northeast win...
9	2024-11-24	Sunday Night	50	Clear	0	2 mph	ENE	False	Clear, with a low around 50.
10	2024-11-25	Monday	78	Sunny	0	2 mph	ENE	True	Sunny, with a high near 78.
11	2024-11-25	Monday Night	55	Clear	0	2 mph	ENE	False	Clear, with a low around 55.
12	2024-11-26	Tuesday	79	Sunny	0	3 mph	NNE	True	Sunny, with a high near 79.
13	2024-11-26	Tuesday Night	58	Clear	0	3 mph	N	False	Clear, with a low around 58.

```

import requests
import pandas as pd
from datetime import datetime, timedelta

# API for Lakeland - chosen rough center of Lakeland Electrical
coverage_map_lat, lon = 28.05, -81.96
points_url = f'https://api.weather.gov/points/{lat},{lon}'
# Define headers for the API
request_headers = {

    "User-Agent": "myweatherapp
(contact@myweatherapp.com)" }
# Fetch the grid information response
response = requests.get(points_url,
headers=request_headers)
if response.status_code == 200:
    data = response.json()
    forecast_hourly_url = data["properties"]["forecastHourly"]
    grid_url = data["properties"]["forecastGridData"]
    print("Hourly Forecast URL:",
forecast_hourly_url)
    print("Grid Forecast URL:", grid_url)
    # Grid data for relative humidity and dew point
    grid_response = requests.get(grid_url, headers=request_headers)
    if

```

```

grid_response.status_code == 200:
grid_data = grid_response.json()
    # Relative Humidity    if
"relativeHumidity" in
grid_data["properties"]:
    rh_data =
pd.DataFrame(grid_data["properties"]["relativeHumidity"]["values"])
rh_data["startDate"] = rh_data["validTime"].str.split("T").str[0]
rh_data["startTime_hour"] = rh_data["validTime"].str.extract(r"T(\d{2})")[0]
rh_data = rh_data[["startDate", "startTime_hour", "value"]]
rh_data.rename(columns={"value": "relativeHumidity"}, inplace=True)
else:

    rh_data = pd.DataFrame(columns=["startDate", "startTime_hour",
"relativeHumidity"])

    # Dew Point (if available)
if "dewpoint" in
grid_data["properties"]:
    dewpoint_data =
pd.DataFrame(grid_data["properties"]["dewpoint"]["values"])
dewpoint_data["startDate"] = dewpoint_data["validTime"].str.split("T").str[0]
    dewpoint_data["startTime_hour"] =
dewpoint_data["validTime"].str.extract(r"T(\d{2})")[0]
dewpoint_data["dewpoint"] = dewpoint_data["value"].apply(
lambda x: round((x * 9 / 5) + 32, 1) if pd.notna(x) else None
)

    dewpoint_data = dewpoint_data[["startDate", "startTime_hour", "dewpoint"]]

else:

    dewpoint_data = pd.DataFrame(columns=["startDate", "startTime_hour",
"dewpoint"])

# Hourly forecast    hourly_response =
requests.get(forecast_hourly_url, headers=headers)    if
hourly_response.status_code == 200:    hourly_data =
hourly_response.json()    hourly_periods =
hourly_data["properties"]["periods"]
    # Convert to DataFrame
hourly_df =
pd.DataFrame(hourly_periods)    #
Precipitation parsing - only output the
number
    hourly_df["probabilityOfPrecipitation"] =
hourly_df["probabilityOfPrecipitation"].apply(
lambda x: x["value"] if x and x["value"] is not None else
None

```

```

    )

    # Split startTime into startDate and startTime
    hourly_df["startDate"] = hourly_df["startTime"].str.split("T").str[0]
    hourly_df["startTime_hour"] =
    hourly_df["startTime"].str.split("T").str[1].str[:2]
    # Filter data for 1-week forecast

    one_week_ahead = (datetime.now() + timedelta(days=7)).strftime("%Y-%m-
%d")
    hourly_df = hourly_df[hourly_df["startDate"] <=
one_week_ahead]
    # Columns to
keep
    hourly_df
= hourly_df[[
    "startDate", "startTime_hour", "temperature", "shortForecast",

    "probabilityOfPrecipitation", "windSpeed", "windDirection"

]]

    # Merge Relative Humidity and Dew Point with Hourly Forecast

    hourly_with_grid = pd.merge(hourly_df, rh_data, on=["startDate",
"startTime_hour"], how="left")

    hourly_with_grid = pd.merge(hourly_with_grid, dewpoint_data,
on=["startDate",
"startTime_hour"], how="left")
    print("\nHourly
Forecast with Relative Humidity and Dew Point:")
    display(hourly_with_grid)
    else:
        print("Error
fetching hourly forecast data")
    else:
        print("Error fetching grid forecast data:",
grid_response.status_code, grid_response.text)
    else:
        print("Error fetching grid information")

Hourly Forecast URL:
https://api.weather.gov/gridpoints/TBW/90,104/forecast/hourly

Grid Forecast URL:
https://api.weather.gov/gridpoints/TBW/90,104 Hourly
Forecast with Relative Humidity and Dew Point:

```

	startDate	startTime_hour	temperature	shortForecast	probabilityOfPrecipitation	windSpeed	windDirection	relativeHumidity	dewpoint
0	2024-11-20	12	77	Chance Showers And Thunderstorms	46	9 mph	S	NaN	NaN
1	2024-11-20	13	79	Chance Showers And Thunderstorms	34	9 mph	S	NaN	67.0
2	2024-11-20	14	79	Chance Showers And Thunderstorms	30	9 mph	SSW	91.0	70.0
3	2024-11-20	15	79	Chance Showers And Thunderstorms	26	9 mph	W	90.0	72.0
4	2024-11-20	16	79	Slight Chance Rain Showers	20	9 mph	WNW	89.0	73.0
...	...	...	...	...	...	...	...	...	...
151	2024-11-26	19	68	Clear	0	2 mph	NNW	53.0	59.0
152	2024-11-26	20	67	Clear	0	2 mph	NNW	52.0	NaN
153	2024-11-26	21	65	Clear	0	2 mph	N	54.0	58.0
154	2024-11-26	22	64	Clear	0	2 mph	N	58.0	NaN
155	2024-11-26	23	62	Clear	0	2 mph	N	65.0	59.0

Dewpoint values are inconsistent - not every hour has that forecast

```
# Filter the DataFrame for rows where dewpoint is not
NaN filtered_data =
hourly_with_grid[hourly_with_grid["dewpoint"].notna()] #
Check if any data exists if not filtered_data.empty:
print("Rows with non-NaN Dewpoint values:")
display(filtered_data) else:
    print("No rows with non-NaN Dewpoint
values.") Rows with non-NaN Dewpoint
values:
```

	startDate	startTime_hour	temperature	shortForecast	probabilityOfPrecipitation	windSpeed	windDirection	relativeHumidity	dewpoint
1	2024-11-20	13	79	Chance Showers And Thunderstorms	34	9 mph	S	NaN	67.0
2	2024-11-20	14	79	Chance Showers And Thunderstorms	30	9 mph	SSW	91.0	70.0
3	2024-11-20	15	79	Chance Showers And Thunderstorms	26	9 mph	W	90.0	72.0
4	2024-11-20	16	79	Slight Chance Rain Showers	20	9 mph	WNW	89.0	73.0
5	2024-11-20	17	76	Slight Chance Rain Showers	20	8 mph	WNW	91.0	74.0
...	...	...	...	...	...	...	...	...	...
146	2024-11-26	14	78	Sunny	0	3 mph	NE	79.0	58.0
147	2024-11-26	15	78	Sunny	0	3 mph	N	72.0	60.0
151	2024-11-26	19	68	Clear	0	2 mph	NNW	53.0	59.0
153	2024-11-26	21	65	Clear	0	2 mph	N	54.0	58.0
155	2024-11-26	23	62	Clear	0	2 mph	N	65.0	59.0

HISTORICAL DATA

- NOAA.gov - <https://www.ncdc.noaa.gov/cdo-web/webservices/v2>

- API:

- Email: jjackson3465@floridapoly.edu

- Token: `idqwrJvPPATpYzveCAWajOVtSQGvFKpR`

- Station info: <https://www.ncdc.noaa.gov/cdo->

- web/datasets/GHCND/locations/FIPS:121

05/detail - Polk County import requests import

pandas as pd from datetime import datetime,

```

timedelta # NOAA CDO API base URL and token
base_url = "https://www.ncdc.noaa.gov/cdo-
web/api/v2/data"
- token = "idqwrjvPPATpYzveCAWajOVtSQGvFKpR" # NOAA API - DON'T
MAKE PUBLIC
-
- # Calculate dates current_date =
datetime.now().strftime("%Y-%m-
%d")
- week_timeframe = (datetime.now() - timedelta(weeks=1)).strftime("%Y-
%m-%d") # Change weeks value to however many you want
- # Define headers
with the token headers
= {"token": token}
- # Function to fetch data for
a specific datatype def
fetch_data(datatype):
- params = {
- "datasetid": "GHCND", # Daily summaries dataset
- "locationid": "FIPS:12105", # Polk County FIPS Code
- "startdate": week_timeframe,
- "enddate": current_date,
- "datatypeid": datatype, # Specific datatype (e.g., TMIN, TMAX, PRCP)
- "limit": 1000,
- "units": "standard"
- }
- response = requests.get(base_url,
headers=headers, params=params) if
response.status_code == 200:
- data =
response.json() return
pd.DataFrame(data.get("results
", [])) else:
- print(f"Error fetching {datatype} data:", response.status_code,
response.text) return pd.DataFrame()
-
- # Fetch data for each datatype tmin_data =
fetch_data("TMIN") tmax_data = fetch_data("TMAX") prcp_data =
fetch_data("PRCP") # Combine all data into a single DataFrame
historical_df = pd.concat([tmin_data, tmax_data, prcp_data],
ignore_index=True)
- # Clean and aggregate the data # Ensure the value
column is numeric historical_df["value"] =
pd.to_numeric(historical_df["value"], errors="coerce") # Pivot
table to create separate columns for TMIN, TMAX, and PRCP

```

```

averages aggregated = historical_df.pivot_table( index="date",
# Group by date
-     columns="datatype",      # Use datatype (TMIN, TMAX,
PRCP) as columns, precipitation measured in inches, temps
measured in F   values="value",      # Aggregate the "value"
column aggfunc="mean"      # Calculate the mean for each
date and datatype
-     ).reset_index()
-     # flatten the column index - for some reason messes up column names if
don't do this aggregated.columns.name = None
-     aggregated.rename(columns={"TMIN": "tempMin", "TMAX": "tempMax",
"PRCP": "precipitation"}, inplace=True)
-     # Make lists of the values averaged to help see what's going on
-     # group by the different datatypes
-     grouped_values = historical_df.groupby(["date",
"datatype"])[["value"]].apply(list).reset_index()
-
-     # pivot so they match
-     lists_pivot = grouped_values.pivot(index="date",
columns="datatype", values="value").reset_index() #
rename columns lists_pivot.columns.name = None
-     lists_pivot.rename(columns={"TMIN": "tempMin_values", "TMAX":
"tempMax_values", "PRCP": "precipitation_values"}, inplace=True)
-     # Merge dfs final_df = pd.merge(aggregated,
lists_pivot, on="date", how="left") # Clean up the
`date` column to keep only the YYYY-MM-DD part
final_df["date"] = final_df["date"].str[:10]
-     # Round the `precipitation` column to 2 decimal
places final_df["precipitation"] =
final_df["precipitation"].round(2) final_df["tempMax"] =
final_df["tempMax"].round(1) final_df["tempMin"] =
final_df["tempMin"].round(1)
-     # output
print("Aggregated
Daily Weather:")
display(final_df)

```

Appendix 6I: Initial EDA Exploration File

[Initial_eda_exploration](#)

Appendix 6J: Random Forest Buckets Python Notebook File

[capstone_random_forrest_buckets.ipynb - Colab](#)